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Observations of the Lyman- α Universe

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Keywords

Ly α emission, galaxy formation, cosmic reionization, cosmology

Abstract

Hydrogen Lyman- α (Ly α) emission has been one of the major observational probes for the high-redshift Universe since the first discoveries of high- z Ly α -emitting galaxies in the late 1990s. Due to the strong Ly α emission originated by resonant scattering and recombination of the most abundant element, Ly α observations witness not only HII regions of star formation and active galactic nuclei (AGNs) but also diffuse HI gas in the circumgalactic medium (CGM) and the intergalactic medium (IGM). Here, we review Ly α sources and present theoretical interpretations reached to date. We conclude the following:

- A typical Ly α emitter (LAE) at $z \gtrsim 2$ with a L^* Ly α luminosity is a high- z counterpart of a local dwarf galaxy, a compact metal-poor star-forming galaxy (SFG) with an approximate stellar (dark matter halo) mass and star-formation rate of $10^{8-9} M_{\odot}$ ($10^{10-11} M_{\odot}$) and $1-10 M_{\odot} \text{ year}^{-1}$, respectively.
- High- z SFGs ubiquitously have a diffuse Ly α -emitting halo in the CGM extending to the halo virial radius and beyond.

- Remaining neutral hydrogen at the epoch of cosmic reionization makes a strong dimming of Ly α emission for galaxies at $z > 6$ that suggests the late reionization history.

The next-generation large-telescope projects will combine Ly α emission data with H I Ly α absorptions and 21-cm radio data that map out the majority of hydrogen (H I+H II) gas, uncovering the exchanges of (*a*) matter by outflow and inflow and (*b*) radiation, relevant to cosmic reionization, between galaxies and the CGM/IGM.

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1. INTRODUCTION

Hydrogen Lyman- α (Ly α) emission is a critical probe for understanding the high-redshift Universe. Due to the abundant hydrogen and the atomic electron transition between the lowest energy levels, from the $n = 2$ state to the ground ($n = 1$) state, Ly α is one of the strongest emission lines produced in the Universe. Although Ly α has the physical characteristics of a resonance line¹ whose photons have more chances of dust attenuation in gaseous nebulae in the process of resonant scattering, the short wavelength (1216 Å) of Ly α emission allows us to pinpoint faint high- z objects by optical and near-infrared (NIR) observations very efficiently. Ly α is an excellent probe of high- z objects near the observational redshift frontier.

¹ Because the chance that electrons stay at the ground state is very high, due to a short timescale required for ground-state transitions, Ly α photons are resonantly scattered by hydrogen.

Historically, Partridge & Peebles (1967) first predicted the importance of a high- z galaxy that emits a strong Ly α line. Partridge & Peebles (1967) argue that an early galaxy produces strong Ly α emission by the recombination process in the interstellar medium (ISM) heated by young massive stars and that up to 6–7% of the total galaxy luminosity can be converted to a Ly α luminosity that is as bright as $\sim 2 \times 10^{45}$ erg s $^{-1}$ at $z \sim 10$ –30. Furthermore, Partridge & Peebles (1967) claim a possibility of strong free-electron scattering in ionized gas of the intergalactic medium (IGM) that smears radiation from young galaxies. Although recent studies do not agree with the extremely bright Ly α luminosity and the notable smearing of the radiation in the ionized IGM, Partridge & Peebles (1967) provide an excellent imaginative theory paper (published more than a half century ago), suggesting that strong Ly α emission of high- z galaxies is physically related to early galaxies and to the ionization state of the IGM; these are two major topics discussed today with regard to strong Ly α emission, namely galaxy formation and cosmic reionization.

After the theoretical predictions of Partridge & Peebles (1967), a number of programs searched for high- z galaxies with strong Ly α emission by observations. However, no such objects were found until the mid-1990s due to the limited sensitivities of the available telescopes. Finally, by the operation starts of large (8–10-m class) ground-based telescopes and the *Hubble Space Telescope* (HST), a few high- z galaxies with strong Ly α emission were successfully identified by narrowband (NB) imaging² and spectroscopy on the sky around a quasi-stellar object (QSO) BR2237-0607 at $z = 4.55$ (Hu & McMahon 1996) and a radio galaxy 53W002 at $z = 2.39$ (Pascarelle et al. 1996), as well as in the blank field (Cowie & Hu 1998). Subsequently, a number of deep observation programs have been conducted for Ly α -emitting objects at $z \gtrsim 2$, including the Hawaii Survey (Cowie & Hu 1998), the Large Area Lyman Alpha Survey (Rhoads et al. 2000), the Subaru surveys (Ouchi et al. 2003), the Multiwavelength Survey by Yale–Chile (Gawiser et al. 2007), the Hobby-Eberly Telescope Dark Energy Experiment (HETDEX) Pilot Survey (Adams et al. 2011), and the Very Large Telescope (VLT)/Multi Unit Spectroscopic Explorer (MUSE) survey (Bacon et al. 2017). Furthermore, Ly α -emitting objects in the local and low- z Universe are investigated by space-based UV observations such as by the HST program of Lyman-Alpha Reference Sample (Östlin et al. 2014) and the *Galaxy Evolution Explorer* (GALEX) grism programs (Deharveng et al. 2008, Cowie et al. 2011). Thus far, Ly α -emitting galaxies with no active galactic nucleus (AGN) have a Ly α luminosity of $\simeq 10^{41}$ – 10^{44} erg s $^{-1}$, which is more than an order of magnitude fainter than that predicted by Partridge & Peebles (1967).

Ly α -emitting objects are called Lyman- α emitters (LAEs). Conventionally, LAEs are defined as objects with a rest-frame Ly α equivalent width (EW) of

$$EW_0 \gtrsim 20 \text{ \AA}. \quad 1.$$

This Ly α EW_0 limit corresponds to that of the samples made by classical observations for LAEs with an NB whose wavelength transmission width is $\sim 1\%$ of the central wavelength.³ It is known that LAEs are young star-forming galaxies (SFGs) or AGNs (Section 3). **Figure 1** is an illustration of a conceptual LAE with spectroscopic properties that are detailed in Section 2. In **Figure 1** and hereafter, we use the abbreviations cMpc, pMpc, and pkpc for comoving megaparsec, physical megaparsec, and physical kiloparsec, respectively.

Because Ly α photons are resonantly scattered by neutral hydrogen H I , Ly α is unfortunately thought to be a poor probe of galaxy dynamics or the ionizing photon budget. However, the distribution and dynamics of H I gas are encoded to Ly α emission via the resonant scattering. The resonantly scattered Ly α photons allow us to investigate H I gas in and around high- z objects

cMpc:
comoving megaparsec

pMpc:
physical megaparsec

pkpc:
physical kiloparsec

²Ly α is redshifted to the NB transmission, showing an NB photometry excess.

³Note that some of the recent studies refer to objects with weak Ly α of $EW_0 > 0 \text{ \AA}$ as LAEs.

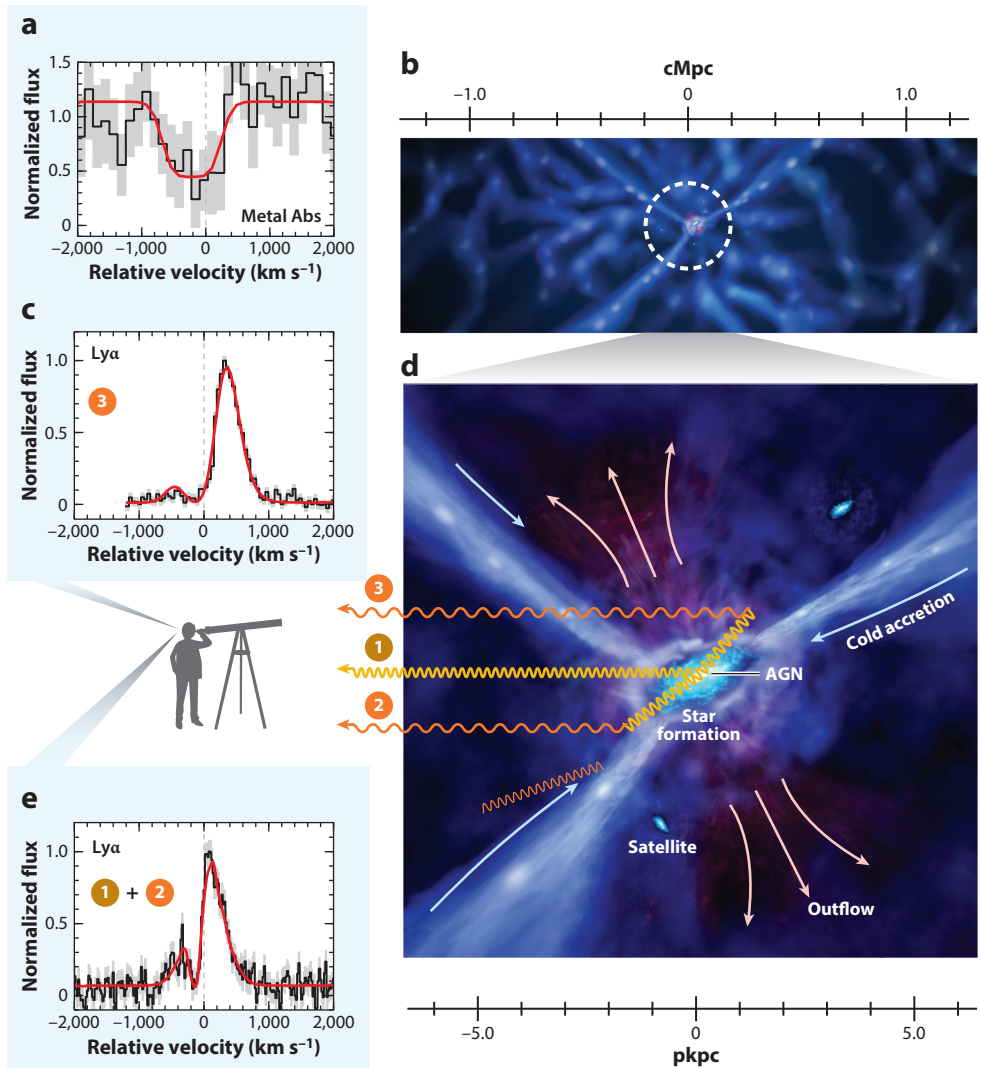


Figure 1

Conceptual figure of an LAE with a moderately high mass. In panel *b*, the LAE is located at the central node of the filamentary structure surrounded with satellite galaxies. The white dashed circle represents the virial radius of the host dark matter halo of the LAE. Panel *d* is a zoom-in of the LAE. Possible origins of Ly α emission are labeled: star formation, AGN, cold accretion, outflow, and satellite galaxies. In the LAE with outflowing gas, the wavy lines (1–3) indicate three light paths for the observed Ly α lines. Panels *e* and *c* present spectra that are dominated by the contributions of the light paths of 1+2 and 3, respectively, with the observed spectrum (black line) and error (gray region) and the best-fit models (red line). The light paths of 2 and 3 indicate that Ly α photons are resonantly scattered in the outflowing gas. In panel *a*, the continuum spectrum of the LAE with a metal absorption line is shown. The spectral data are adapted with permission from Hashimoto et al. (2015) and Sugahara et al. (2019). Abbreviations: AGN, active galactic nucleus; LAE, Lyman- α emitter.

that is generally hard to probe by observations. The circumgalactic medium (CGM) of high- z SFGs is characterized by diffuse Ly α emission extending over >10 pkpc even to the large scale reaching the IGM (Kakuma et al. 2019; Section 7). Interestingly, such diffuse Ly α emission of high- z galaxies covers the entire sky in any observational lines of sight (Wisotzki et al. 2018). **Figure 1b,d** illustrates Ly α emission traveling in the CGM and IGM. Theoretical models predict that some fractions of Ly α photons are produced by cold accretion, outflow, and unresolved faint satellites (Section 2) especially in the CGM and IGM. At the epoch of reionization (EoR), the ionized fraction and distribution of H I in the IGM are imprinted in the observed Ly α emission (Section 9). Ly α is an important probe for gas of the CGM and the IGM for characterizing galaxies and cosmic reionization.

2. PHYSICAL PICTURE OF LYMAN- α EMITTERS

2.1. Origins of Ly α Emission

Ly α emission from galaxies has five major origins. In the ISM near the central region of a galaxy, Ly α emission can originate from recombinations of hydrogen atoms that are ionized mainly by two sources: (a) young massive stars (star formation) and (b) an AGN if any. In the CGM and the outer region, Ly α can be emitted from three origins: (a) outflow that can collisionally excite H I atoms (shock heating);⁴ (b) infalling gas (cold accretion), which is predicted to release a significant amount of its gravitational energy in Ly α through collisional excitation (gravitational cooling); and (c) fluorescence from hydrogen in the CGM and IGM photoionized by UV background radiation supplied by energetic sources such as QSOs. In addition, star formation in unresolved faint satellite galaxies can also contribute. Note that no Ly α sources dominated by gravitational cooling have been definitively identified so far (Section 7).

Another important physical process for Ly α emission is resonant scattering, because H I gas in typical galaxies is optically thick to Ly α . The cross section of Ly α for a collection of moving atoms can be obtained by convolving the single-atom cross section with their velocity distribution. Assuming a Maxwellian velocity distribution and introducing the dimensionless frequency variable $x \equiv (\nu - \nu_0)/\Delta\nu_D$, where ν_0 is the line center frequency and $\Delta\nu_D$ is the Doppler width, the average cross section is⁵

$$\sigma_x(\nu, T) = \frac{3\lambda_0^2 a_v}{2\sqrt{\pi}} H(a_v, x) \simeq 5.9 \times 10^{-14} \left(\frac{T}{10^4 \text{ K}} \right)^{-1/2} H(a_v, x) \text{ cm}^2, \quad 2.$$

where λ_0 is the Ly α wavelength, a_v is the Voigt parameter, and $H(a_v, x)$ is the Voigt function,

$$H(a_v, x) = \frac{a_v}{\pi} \int_{-\infty}^{\infty} \frac{e^{-y^2} dy}{(y-x)^2 + a_v^2} \approx \begin{cases} e^{-x^2} & \text{central resonant core,} \\ \frac{a_v}{\sqrt{\pi} x^2} & \text{damping wing.} \end{cases} \quad 3.$$

The transition between the core and wing happens at around $e^{-x^2} = a_v/(\sqrt{\pi}x^2)$. Based on these equations, H I gas is optically thick to Ly α at the line center when the H I column density is higher than $N_{\text{HI}} = 1/\sigma_x(\nu_0, T) \simeq 2 \times 10^{13} (T/10^4 \text{ K})^{1/2} \text{ cm}^{-2}$, which is a measurement much lower than those of typical galaxies. Although the resonant nature of Ly α makes it difficult to pinpoint the original Ly α source position, it enables investigations of the distribution and the kinematics of H I gas via theoretical modeling.

⁴In the case of collisional excitation, the Ly α -to-H α flux ratio can be higher than that of the case B recombination (figure 1.7 of Dijkstra 2019).

⁵A detailed derivation of this equation is given in section 1.5.4 of Dijkstra (2019).

2.2. Modeling Ly α Emission

One of the key issues to modeling Ly α emission from galaxies is its complicated radiative transfer due to the resonant nature of Ly α .⁶ Analytical solutions have been obtained only in very limited cases. In the simple case of an optically thick dust-free static slab with a central plane source emitting Ly α , the emergent Ly α line profile is given by Harrington (1973)⁷ as

$$J(x) = \frac{\sqrt{6}}{24\sqrt{\pi}a_v\tau_0} \frac{x^2}{\cosh\left(\sqrt{\frac{\pi^3}{54}} \frac{x^3}{a_v\tau_0}\right)}, \quad 4.$$

where τ_0 is the optical depth at the line center from the center to the boundary of the slab. The spectral shape is symmetric around $x = 0$ and double peaked at $x \approx \pm 1.1(a_v\tau_0)^{1/3}$. The peaks are more separated with higher τ_0 , because Ly α photons need to shift their velocities more into the wings to escape.

For more general cases, there are some theoretical models, such as the expanding shell (ES) model (Ahn 2004, Verhamme et al. 2006), that adopt the Monte Carlo radiative transfer technique, and these models successfully explain the diversity of observed Ly α profiles with a relatively small number of physical parameters. The ES model assumes a simple geometry in which a Ly α source is located at the center of a spherically symmetric ES of homogeneous and isothermal H I gas, modeling a galaxy-scale supershell of outflow made by multiple supernovae in star-forming regions⁸ (Figure 1d).

To explain the basic idea of the ES model, Figure 1 shows the predictions of observed Ly α emission line profiles for three different light paths (①–③). Ly α photons along the light path (①) directly come from the central Ly α source penetrating the shell. Along the light path (②), Ly α photons escape from the shell approaching the observer via scattering, although some of them are absorbed by H I in the shell, resulting in a red component escaping the red wing of the Ly α absorption in the blueshifted shell and a small blue component escaping the blue wing of Ly α . Figure 1e shows the Ly α spectrum that is dominated by these two components (①+②). The velocity shift of the red component relative to the systemic redshift is small, because a small number of Ly α photons escape from the far side of the shell blowing away by scattering that is referred to as backscattering. Ly α photons tracking the light path (③) experience the backscattering at the shell. In this case, the Ly α spectrum can have a peak at $\sim 2V_{\text{exp}}$, where V_{exp} is the radial expansion velocity or the outflow velocity (see Figure 1c). The $\sim 2V_{\text{exp}}$ shift of the peak is given by an effect similar to that of a reflection in a moving mirror; the Ly α photons enter the shell experiencing a Doppler shift by V_{exp} and are then backscattered to the observer being further Doppler shifted by V_{exp} . This simple model has successfully reproduced observed Ly α profiles of LAEs as well as of Lyman break galaxies (LBGs) at both low and high redshifts (Section 6).

Some theoretical studies have conducted cosmological hydrodynamic simulations combining Monte Carlo radiative transfer calculations with dust attenuation effects to investigate possible descendants of high- z LAEs as well as their physical properties. Yajima et al. (2012) have found that progenitors of a present-day Milky Way (MW)-like galaxy with a dark matter halo mass $\sim 10^{12} M_{\odot}$ show bright Ly α emission at high redshifts comparable with Ly α L^* ($L_{\text{Ly}\alpha}^*$) of LAEs at $z \sim 2$ –6. Figure 2a shows the distributions of gas and stars of the most massive progenitor of a MW-like galaxy as well as the Ly α surface brightness (SB) at different redshifts. Figure 2b

⁶In other words, Ly α photons scatter multiple times in the optically thick line core before diffusing into the wings, where they finally escape.

⁷See also equation 3.51 and the footnote on pg. 29 of Laursen (2010).

⁸Such galaxy-scale expanding supershells are found in nearby starbursts (Marlowe et al. 1995).

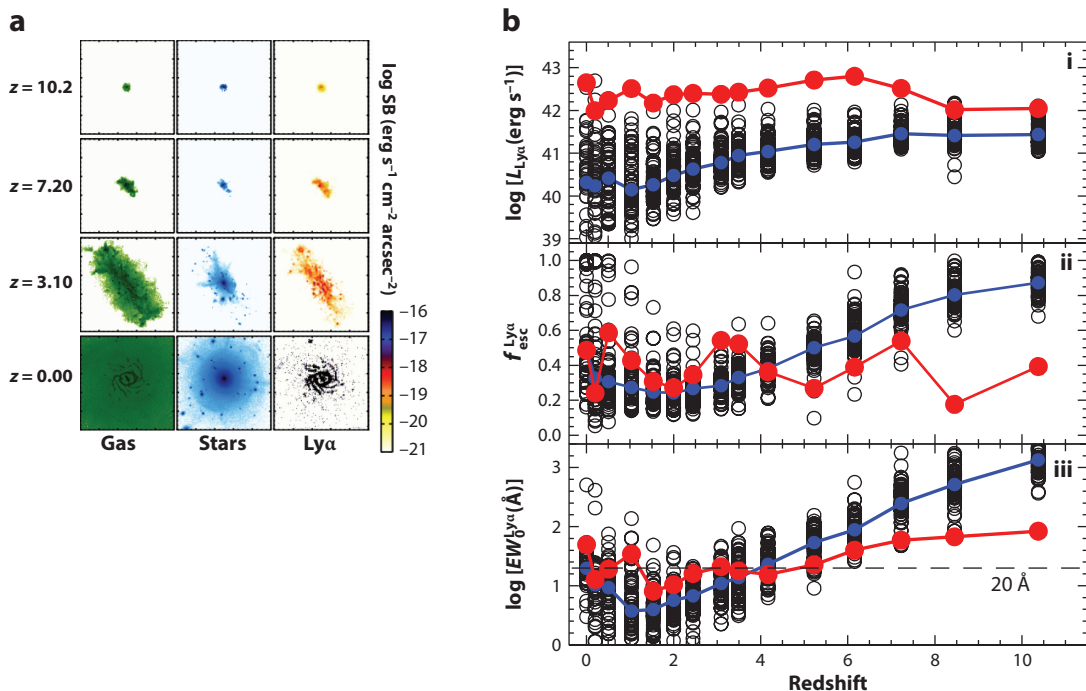


Figure 2

(a) Snapshots of a Milky Way-like galaxy at different redshifts in a cosmological hydrodynamic simulation. From left to right, the distributions of gas and stars as well as the SB of Ly α are shown. The box size is 200 pkpc. (b) Evolution of the Ly α properties of the 60 most massive progenitors. The subpanels present (b, i) the Ly α luminosity, (b, ii) the escape fraction of Ly α photons over a whole solid angle, and (b, iii) Ly α EW_0 . The red-filled circles correspond to the most massive progenitor at each redshift. The blue-filled circles represent the median value of the 60 galaxies. Note that observational effects of reionization, Ly α dimming in the IGM (Section 9), are not included in the models. Adapted from Yajima et al. (2012) with permission. Abbreviations: EW_0 , rest-frame equivalent width; pkpc, physical kiloparsec; SB, surface brightness.

represents the evolution of Ly α properties of the 60 most massive progenitors. As shown in **Figure 2b, iii**, most of the galaxies at high redshifts are classified as LAEs, and some of them are bright enough to be detected in previous observations (see **Figure 2b, i**). The escape fractions of Ly α photons over a whole solid angle are also presented in **Figure 2b, ii**. The median value of the Ly α escape fraction is about 30% at $z \sim 3$, which is consistent with those obtained in observational studies (Section 3). Note that at low redshifts of $z \lesssim 2$, the Ly α escape fraction has a relatively large dispersion due to the diversities in their physical properties. These results suggest that some of the typical observed LAEs at $z \sim 2$ –6 would evolve into present-day MW-like galaxies, which is consistent with the results from the clustering analyses (Gawiser et al. 2007, Ouchi et al. 2010).

3. LYMAN- α EMITTER OBSERVATIONS

As introduced in Section 1, LAEs have been identified by many observational studies. The successful classical technique to find LAEs is NB imaging. In this technique, LAEs are selected on the basis of their NB excesses compared to broadbands (**Figure 3**). Imaging surveys with wide-field cameras such as Subaru's Suprime-Cam have constructed large samples of LAEs, and their follow-up spectroscopic campaigns have confirmed the validity of the NB imaging technique with low fractions of contaminants. The recent advent of new wide-field cameras including Hyper

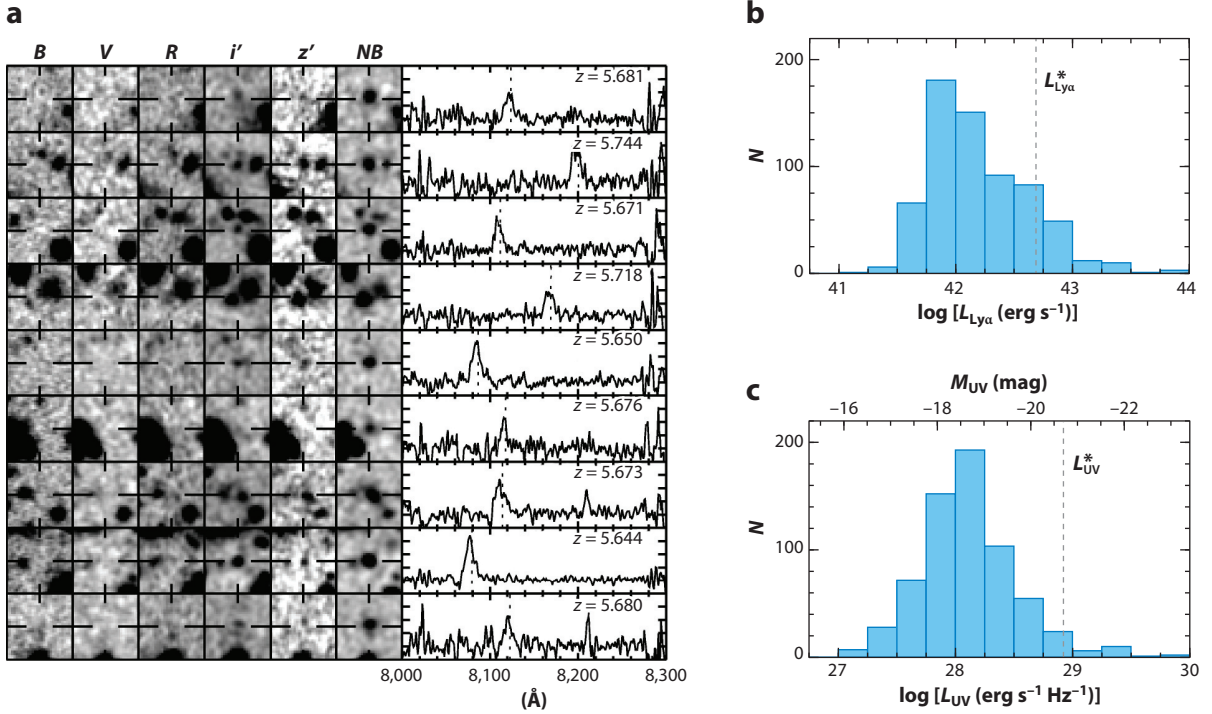


Figure 3

(a) Examples of cutout images and spectra of LAEs residing at $z = 5.7$. The size of each cutout is 6 arcsec $\times$ 6 arcsec. In the spectrum panels, the vertical dashed line is the central wavelength of the detected emission line. (b) Histogram of the Ly α luminosities of $z \sim 2$ –3 LAEs compiled by Shibuya et al. (2019). The characteristic Ly α luminosity $L_{\text{Ly}\alpha}^*$ at $z = 2.2$ (see Equation 5 and the text) is presented with a vertical dashed line. (c) Same as panel b, but for UV luminosity. The vertical dashed line is the characteristic UV luminosity L_{UV}^* at $z \sim 2$. Panel a adapted with permission from Ouchi et al. (2008) with permission. Abbreviations: LAE, Lyman- α emitter; NB, narrowband.

Suprime-Cam (HSC) now allows LAE surveys over cosmological volumes (~ 0.5 comoving Gpc² area; Ouchi et al. 2018).

Another successful technique is blind spectroscopy. In particular, the integral field spectrograph of VLT/MUSE has recently yielded complementary results to those of NB imaging. Thanks to the wider wavelength coverage and higher spectral resolution, MUSE is suitable in searches for LAEs over a wide redshift range down to a flux limit fainter than the NB observations (Drake et al. 2017), although the field of view of MUSE is small. Another remarkable effort is that by a wide-field fiber spectrograph survey, HETDEX (Hill et al. 2008), which covers a ~ 400 -deg² sky via a blind search for relatively bright LAEs.

Because strong Ly α emission from high- z galaxies is redshifted in a wavelength range of optical and NIR in which deep spectroscopic data can be obtained, Ly α emission has been detected in many galaxies by spectroscopy at the high-redshift frontier. **Table 1** presents the list of galaxies identified, to date, at a spectroscopic redshift of $z_{\text{spec}} > 7.2$. Some high- z galaxies are found only with the Ly α continuum break (Oesch et al. 2016) or [OIII] 88- μ m emission (Tamura et al. 2019). Previously it was thought that Ly α photons cannot escape from galaxies especially at the early EoR due to the resonant scattering effect in the neutral IGM. However, as summarized in **Figure 4**, many galaxies found to date show Ly α in emission even at $z \sim 9$ beyond the heart of reionization,

Table 1 List of spectroscopically identified galaxies at $z_{\text{spec}} > 7.2$

Object ID	Right ascension (J2000)	Declination (J2000)	Spectroscopic redshift z_{spec}	Intrinsic UV abs. magnit. M_{UV}	$\text{Ly}\alpha$ EW_0^a (Å)	Probe	Other lines	Reference
GN-z11	12:36:25.46	+62:14:31.4	11.09	-22.1 ± 0.2	ND	Lyman break	ND	Oesch et al. 2016
MACS11149-JD	11:49:33.58	+22:24:45.7	9.1096	−19.0	11.4	[OIII] 88 μm	$\text{Ly}\alpha$	Hashimoto et al. 2018, Hoag et al. 2018
EGSY- 2008532660	14:20:08.50	+52:53:26.60	8.683	−22.0	28	$\text{Ly}\alpha$	ND	Zitrin et al. 2015
A2744-YD4	00:14:24.9	−30:22:56.1	8.382	−20.3	10.7 ± 2.7	[OIII] 88 μm	$\text{Ly}\alpha$	Laporte et al. 2017
MACS0416-Y1	04:16:09.40	−24:05:35.5	8.3118	−20.8	ND	[OIII] 88 μm	ND	Tamura et al. 2019
EGS-zs8-1	14:20:34.89	+53:00:15.4	7.7302	-22.06 ± 0.05	21 ± 4	$\text{Ly}\alpha$	CIII] 1908	Oesch et al. 2015, Stark et al. 2017
z7-GSD-3811	03:32:32.03	−27:45:37.1	7.6637	$-21.22^{+0.06}_{-0.10}$	$15.6^{+5.9}_{-3.6}$	$\text{Ly}\alpha$	ND	Song et al. 2016
MACS1423-z7p64	14:23:46.18	+24:04:10.76	7.640	-19.6 ± 0.2	9 ± 2	$\text{Ly}\alpha$	ND	Hoag et al. 2017
z7-GND-16863	12:37:19.94	+62:15:26.05	7.599	−21.24	61.28 ± 5.85	$\text{Ly}\alpha$	ND	Jung et al. 2019
z8-GND-5296	12:36:37.90	+62:18:08.5	7.506	−21.2	33.19 ± 3.20	$\text{Ly}\alpha$	CIII]1908	Finkelstein et al. 2013, Hu et al. 2019, Jung et al. 2019
EGS-zs8-2	14:20:12.09	+53:00:26.97	7.4770	−21.9	20.2	$\text{Ly}\alpha$	ND	Roberts-Borsani et al. 2016, Stark et al. 2017
GS2-1406	03:33:09.14	−27:51:55.47	7.452	−19.9	140.3 ± 19.0	$\text{Ly}\alpha$	ND	Larson et al. 2018
SDF-NB1006-2	13:24:35.418	+27:27:27.81	7.288	-23.79 ± 0.04	1.99 ± 0.37	$\text{Ly}\alpha$	ND	Shibuya et al. 2012
SXDF-NB1006-2	02:18:56.536	−05:19:58.87	7.215	-21.52 ± 0.18	32	$\text{Ly}\alpha$	[OIII] 88 μm	Shibuya et al. 2012, Inoue et al. 2016
GN-108036	12:36:22.68	+62:08:07	7.213	−21.8	33	$\text{Ly}\alpha$	ND	Ono et al. 2012

Abbreviations: EW_0 , rest-frame equivalent width; ND, no detection or no data.

^a EW in the rest-frame.

Table 2 Schechter parameters for Lyman- α luminosity functions

Redshift	$L_{\text{Ly}\alpha}^*$ (10^{42} erg s $^{-1}$)	$\phi_{\text{Ly}\alpha}^*$ (10^{-4} Mpc $^{-3}$)	α	References
0.3	$0.71^{+0.32}_{-0.29}$	$1.12^{+2.45}_{-0.61}$	−1.8 (fixed)	Cowie et al. 2010, Konno et al. 2016
2.2	$4.87^{+0.83}_{-0.68}$	$3.37^{+0.80}_{-0.66}$	−1.8 (fixed)	Konno et al. 2016
3.1	$8.49^{+1.65}_{-1.46}$	$3.90^{+1.27}_{-0.90}$	−1.8 (fixed)	Ouchi et al. 2008, Konno et al. 2016
3.7	$9.16^{+2.03}_{-1.67}$	$3.31^{+1.42}_{-0.98}$	−1.8 (fixed)	Ouchi et al. 2008, Konno et al. 2016
5.7	$9.09^{+3.67}_{-2.70}$	$4.44^{+4.04}_{-2.05}$	−1.8 (fixed)	Ouchi et al. 2008, Konno et al. 2016
5.7	$16.4^{+21.6}_{-6.2}$	$0.849^{+1.87}_{-0.771}$	$-2.56^{+0.53}_{-0.43}$	Konno et al. 2018
6.6	$16.6^{+3.0}_{-6.9}$	$0.467^{+1.44}_{-0.442}$	$-2.49^{+0.50}_{-0.50}$	Konno et al. 2018
7.0	$15.0^{+4.2}_{-3.1}$	$0.45^{+0.26}_{-0.18}$	−2.5 (fixed)	Itoh et al. 2018
7.3	$5.5^{+94.5}_{-3.3}$	$0.94^{+12.03}_{-0.93}$	−2.5 (fixed)	Konno et al. 2014, Itoh et al. 2018

suggesting a patchy nature of reionization in which relatively luminous ionizing sources such as the spectroscopically confirmed galaxies are located in large ionized bubbles in the IGM, which allow Ly α photons to escape from the neutral IGM.

In total, thanks to the successful selection techniques and intensive spectroscopic follow-up campaigns, to date >1,000 (>20,000) LAEs have been spectroscopically identified (photometrically selected) in the literature (e.g., Drake et al. 2017, Sobral et al. 2018, Shibuya et al. 2019). As shown in **Figure 3c**, LAEs tend to have faint sub- L_{UV}^* ($\sim 0.1L_{\text{UV}}^*$) luminosities, i.e., UV-continuum⁹ luminosities fainter than the characteristic luminosity L_{UV}^* of SFGs (Reddy & Steidel 2009) at similar redshifts by about an order of magnitude. The numerous LAEs with faint UV continua can be understood by the observational fact that a UV-continuum faint SFG has a Ly α -emitting galaxy fraction higher than that of a UV-continuum bright SFG (**Figure 5**), which is known as the Ando effect (Ando et al. 2006).

One of the most fundamental observational quantities to characterize galaxy properties is the luminosity function (LF). **Figure 6a,c** compiles Ly α LF measurements for LAEs over a wide redshift range of $z \sim 0.3$ –7.3. The Ly α LF is often parameterized with a Schechter function (Schechter 1976),

$$\phi(L_{\text{Ly}\alpha})dL_{\text{Ly}\alpha} = \phi_{\text{Ly}\alpha}^* \left(\frac{L_{\text{Ly}\alpha}}{L_{\text{Ly}\alpha}^*} \right)^\alpha \exp \left(-\frac{L_{\text{Ly}\alpha}}{L_{\text{Ly}\alpha}^*} \right) d \left(\frac{L_{\text{Ly}\alpha}}{L_{\text{Ly}\alpha}^*} \right), \quad 5.$$

where $L_{\text{Ly}\alpha}$ is the observed Ly α luminosity, $L_{\text{Ly}\alpha}^*$ is the characteristic Ly α luminosity, $\phi_{\text{Ly}\alpha}^*$ is the normalization, and α is the faint-end slope. A Schechter function is also expressed with a Ly α luminosity in the logarithmic form,

$$\Phi(\log L_{\text{Ly}\alpha}) = (\ln 10) \phi_{\text{Ly}\alpha}^* 10^{(\alpha+1)(\log L_{\text{Ly}\alpha} - \log L_{\text{Ly}\alpha}^*)} \exp \left[-10(\log L_{\text{Ly}\alpha} - \log L_{\text{Ly}\alpha}^*) \right]. \quad 6.$$

The best-fit Schechter functions derived in the literature are also plotted in **Figure 6a,c** and their best-fit Schechter parameters are summarized in **Table 2**. **Figure 6b,d** shows the 1σ and 2σ confidence intervals for the combinations of the Schechter parameters of $L_{\text{Ly}\alpha}^*$ and $\phi_{\text{Ly}\alpha}^*$, where the α values are fixed at fiducial values of −1.8 for low redshifts (**Figure 6b**) and −2.5 for high redshifts (**Figure 6d**).

Ly α LFs show three evolutionary trends: a monotonic increase from $z \sim 0$ to $z \sim 3$, no evolution from $z \sim 3$ to $z \sim 6$, and a rapid drop beyond $z \sim 6$. First, at $z \sim 0$ –3, the Ly α LFs show a strong

⁹Throughout this review, the UV continuum indicates the continuum emission at the rest frame to be $\simeq 1500$ Å. No extinction corrections are applied unless otherwise specified.

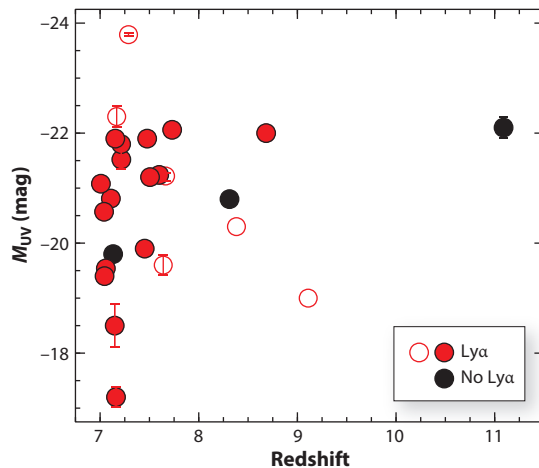


Figure 4

UV absolute magnitude, M_{UV} , as a function of redshift for spectroscopically confirmed galaxies at $z_{\text{spec}} > 7$ taken from **Table 1** and the literature. The red and black circles represent galaxies with and without Ly α detection, respectively. In particular, galaxies with Ly α $EW_0 > 20 \text{ \AA}$ are shown by the red-filled circles. Abbreviation: EW_0 , rest-frame equivalent width.

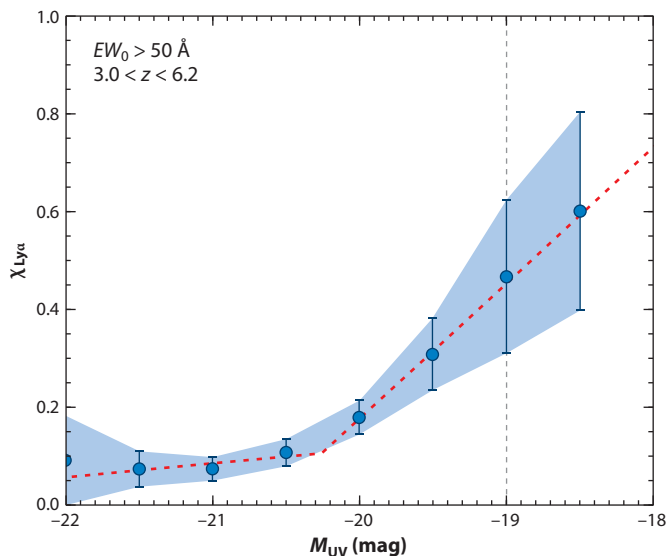


Figure 5

Fraction of spectroscopically confirmed galaxies at $z = 3.0\text{--}6.2$ emitting strong Ly α lines (Ly α $EW_0 > 50 \text{ \AA}$) as a function of M_{UV} . The black vertical dashed line denotes the $\simeq 90\%$ detection completeness limit. The red-dashed lines are the best-fit lines of the first-order polynomials in the ranges of $-22.0 < M_{\text{UV}} < -20.5$ and $-20.0 < M_{\text{UV}} < -18.5$. Figure adapted from Stark et al. (2010) with permission. Abbreviation: EW_0 , rest-frame equivalent width.

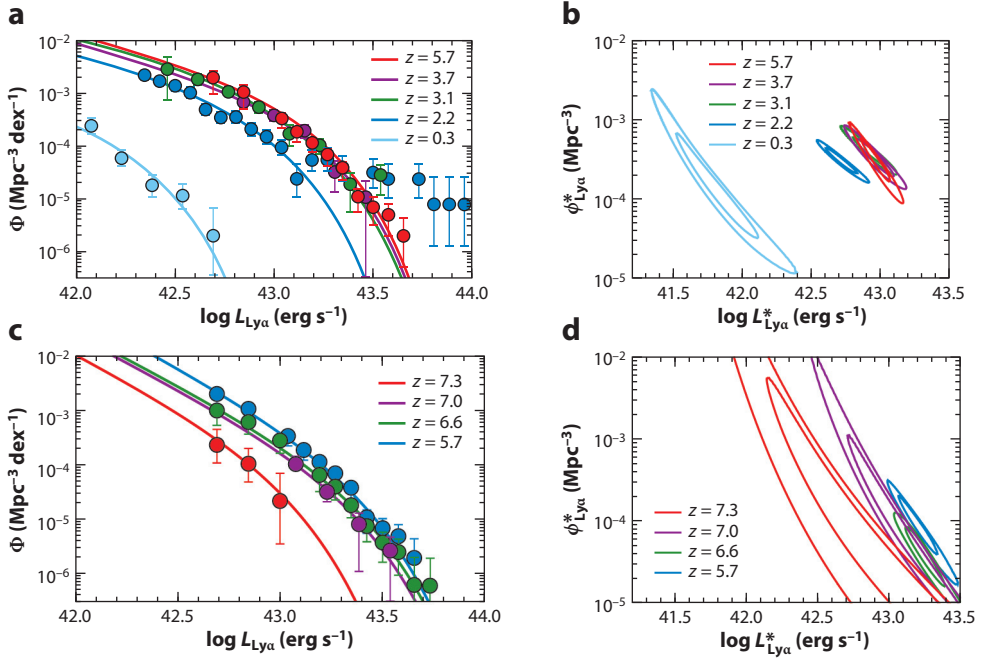


Figure 6

Ly α luminosity functions from (a) $z = 0.3$ to $z = 5.7$ and (c) $z = 5.7$ to $z = 7.3$. The colored curves are their best-fit Schechter functions. Shown are the (b) 1σ - and (d) 2σ -level error contours of their Schechter parameters, $L_{\text{Ly}\alpha}^*$ and $\phi_{\text{Ly}\alpha}^*$. The references of the data are summarized in Table 2.

increase with increasing redshift (Deharveng et al. 2008, Cowie et al. 2010). The rise of Ly α LFs is much larger than that of UV-continuum LFs based on the comparisons of Ly α luminosity densities (LDs); i.e.,

$$\rho_{\text{Ly}\alpha} = \int_{L_{\text{Ly}\alpha}^{\text{lim}}}^{\infty} L_{\text{Ly}\alpha} \phi_{\text{Ly}\alpha}(L_{\text{Ly}\alpha}) dL_{\text{Ly}\alpha}, \quad (7)$$

where $L_{\text{Ly}\alpha}^{\text{lim}}$ is the limiting Ly α luminosity, and the UV-continuum LDs,

$$\rho_{\text{UV}} = \int_{L_{\text{UV}}^{\text{lim}}}^{\infty} L_{\text{UV}} \phi_{\text{UV}}(L_{\text{UV}}) dL_{\text{UV}}, \quad (8)$$

where ϕ_{UV} , L_{UV} , and $L_{\text{UV}}^{\text{lim}}$ are the UV-continuum LF, the UV-continuum luminosity, and the limiting UV-continuum luminosity, respectively. **Figure 7a** presents a comparison of the Ly α and UV-continuum LD evolution and indicates that the rise of the Ly α LD is larger than that of the UV LD from $z \sim 0$ to $z \sim 3$. Second, from $z \sim 3$ to $z \sim 6$, the Ly α LFs show no significant evolution (Ouchi et al. 2008), whereas the UV LFs decrease strongly in the volume number density at $M_{\text{UV}} = -21$ mag by a factor of about 5 (Bouwens et al. 2015). These differences of the evolutionary trends can be explained by the fact that the escape fraction of Ly α photons and/or the ionizing photon production rate from galaxies increase with increasing redshift from $z \sim 0$ to $z \sim 6$ (Hayes et al. 2011; see Section 6 for more details). The Ly α escape fraction is defined as

$$f_{\text{esc}}^{\text{Ly}\alpha} = \frac{L_{\text{Ly}\alpha}}{L_{\text{Ly}\alpha}^{\text{int}}}, \quad (9)$$

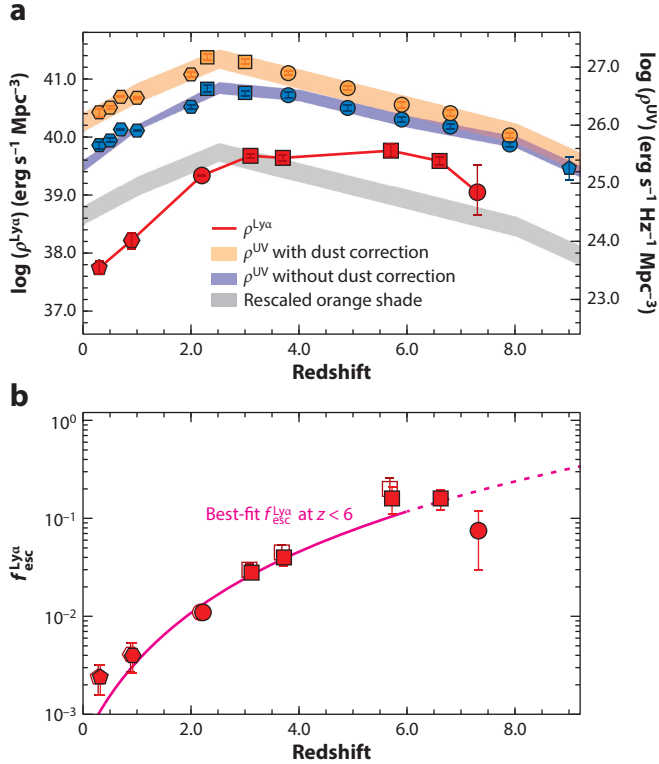


Figure 7

(a) Ly α LDs and UV LDs as a function of redshift. The red symbols indicate Ly α LDs. The orange (blue) symbols and shaded area denote the UV LDs and the errors, respectively, corrected for (no) dust extinction. The gray area is the evolutionary trend of the dust-corrected UV LDs scaled to the Ly α LD at $z \sim 3$ for comparison. (b) Cosmic average Ly α escape fraction, $f_{\text{esc}}^{\text{Ly}\alpha}$, as a function of redshift. The red-filled (open) symbols are $f_{\text{esc}}^{\text{Ly}\alpha}$ estimated from the observed Ly α LDs and the dust-corrected UV LDs without (with) consideration of the effect of intergalactic medium absorption. The magenta solid line represents the best-fit function for the Ly α escape fraction evolution from $z = 0$ to 6. Figure adapted from Konno et al. (2016) with permission. Abbreviation: LD, luminosity density.

where $L_{\text{Ly}\alpha}^{\text{int}}$ is the intrinsic Ly α luminosity that can be estimated from a star-formation rate (SFR), $L_{\text{Ly}\alpha}^{\text{int}} \text{ (erg s}^{-1}) = 1.1 \times 10^{42} \text{ SFR (M}_{\odot} \text{ year}^{-1})$, under the assumption of the case B recombination (footnote 11 of Henry et al. 2015) and the relation between the H α luminosity and SFR (i.e., a constant ionizing photon production rate; Kennicutt 1998). **Figure 7b** presents evolution of the cosmic average $f_{\text{esc}}^{\text{Ly}\alpha}$ values that are estimated with the Ly α and UV LDs via Equation 9. The cosmic average $f_{\text{esc}}^{\text{Ly}\alpha}$ monotonically increases by two orders of magnitude with increasing redshift from $z \sim 0$ to $z \sim 6$. Lastly, the Ly α LD appears to drop faster than the UV LD from $z \sim 6$ to a higher redshift (**Figure 7**). Because this evolutionary trend at $z \gtrsim 6$ is closely related to cosmic reionization, we discuss this evolutionary trend thoroughly in Section 9.

Another notable feature in Ly α LF is the shape of the bright end. At $z \sim 2$ –3, the significant bright-end LF excesses beyond the Schechter functions are found and explained by AGNs by the multiwavelength analysis (Konno et al. 2016) and spectroscopy (Sobral et al. 2018). At redshifts higher than $z \sim 2$ –3, such bright-end LF excess features are not clearly found, probably because the number densities of AGNs decrease with increasing redshift. Interestingly, at the EoR $z \sim 7$,

some studies have reported possible bright-end LF excess detections, arguing that bright LAEs are presumably surrounded by large ionized bubbles (Matthee et al. 2015, Zheng et al. 2017). However, with its great statistical accuracy, the Subaru HSC survey has recently claimed that the Ly α LF at $z \sim 7$ can be explained by the Schechter functional form with no significant bright-end LF excess but rather with a steep slope of $\alpha \sim -2.5$ (Itoh et al. 2018, Konno et al. 2018). The shape of the bright-end LF is under debate (Hu et al. 2019).

Although the previous studies have successfully characterized the overall evolution of Ly α LFs, there is still a fundamental but important open question: What is the lowest mass of dark matter halos of SFGs that can emit Ly α ? The previous observations are not deep enough to detect very faint LAEs. In fact, as can be seen in **Table 2**, most of the previous studies do not obtain good constraints on the faint-end slope α and thus have fixed α at a fiducial value in the Schechter function fitting. From the theoretical point of view, the LF is expected to have a turnover at a faint luminosity because it is difficult for very low-mass dark matter halos to host SFGs owing to inefficiency of gas cooling (Liu et al. 2016).

4. MORPHOLOGICAL PROPERTIES

Since the early discovery of LAEs, we have known of their compact nature from HST observations (Pascarelle et al. 1996). Deep HST extragalactic legacy data have indicated that LAEs have half-light radii of $r_e \sim 1$ pkpc on average in the rest-frame UV and optical stellar continua (Malhotra et al. 2012, Paulino-Afonso et al. 2018, Shibuya et al. 2019). Although substructures are found in some cases, the main stellar components of LAEs basically have a disk-like radial SB profile with a Sérsic index of $n_s \sim 1$ (Taniguchi et al. 2009, Gronwall et al. 2011), which is similar to LBGs (**Figure 8a**). This typical radial SB profile of LAEs has not largely changed over cosmic time similar to other SFG populations (Paulino-Afonso et al. 2018, Shibuya et al. 2019).

Early HST studies have reported that the UV morphology of LAEs shows nearly no r_e evolution over cosmic time (Malhotra et al. 2012, Paulino-Afonso et al. 2018). However, it is claimed that the no r_e evolution results may be obtained by the bias raised by heterogeneous LAE samples in luminosity over the redshift range (Shibuya et al. 2019). Note that the UV-continuum morphology of LAEs follows the r_e -luminosity relation similar to that of LBGs. The r_e -luminosity relation indicates that faint continuum LAEs have a small r_e (Leclercq et al. 2017, Shibuya et al. 2019). If sample selections at different redshifts are not well controlled, observers identify many faint LAEs at low z that have a small r_e , which diminishes a trend of r_e evolution. Comparing bias-controlled and uncontrolled samples, Shibuya et al. (2019) find that the no r_e evolution of LAEs is mistakenly concluded due to the existence of the r_e -luminosity relation and the sample bias. With the bias-controlled samples, the median r_e values of LAEs monotonically evolve as $r_e \sim (1+z)^{-1.37}$, which is similar to those of SFGs and LBGs at a given UV-continuum luminosity (Shibuya et al. 2019; **Figure 8b**).

The spatial offset between Ly α and stellar continuum emission peaks, referred to as the Ly α spatial offset or $\delta_{\text{Ly}\alpha}$, is an important clue to physical properties of LAEs (Jiang et al. 2013, Ouchi et al. 2013). Statistical studies suggest that LAEs with a low Ly α EW_0 typically have a large $\delta_{\text{Ly}\alpha}$ value (Shibuya et al. 2014a, Hoag et al. 2019). This Ly α EW_0 - $\delta_{\text{Ly}\alpha}$ anticorrelation implies that Ly α photons are selectively attenuated by dust due to the long mean-free path of Ly α photons by resonant scattering in abundant H I gas of low Ly α EW_0 galaxies that makes large $\delta_{\text{Ly}\alpha}$ values.

In summary, the disk-like radial SB profile and the r_e values indicate that LAEs have stellar components similar to that of LBGs at a given UV-continuum luminosity. This morphological similarity suggests that the Ly α escape is not strongly related to the morphology of stellar components but instead is governed by the column density, geometry, kinematics, and/or ionization states of the ISM and CGM. This interpretation is also supported by the $\delta_{\text{Ly}\alpha}$ measurements.

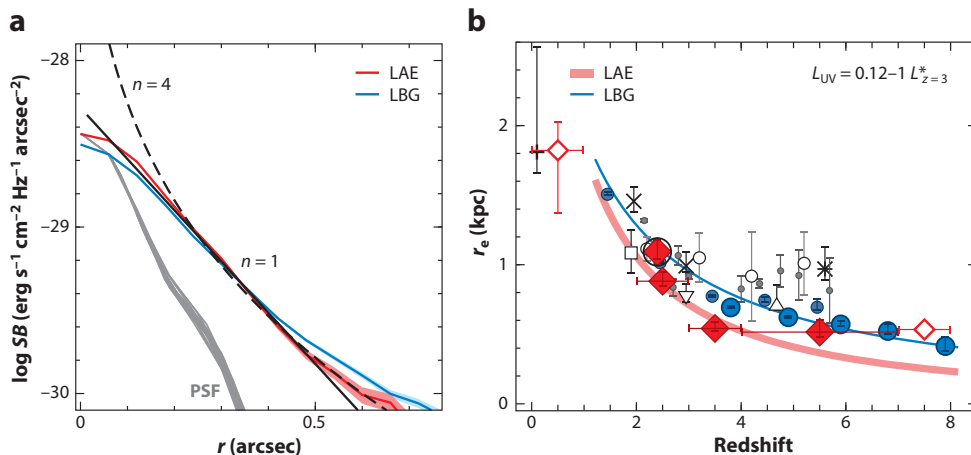


Figure 8

(a) Radial SB profiles of continuum in the rest-frame UV wavelength for LAEs (red curve) and LBGs (blue curve) at $z \sim 3-4$ (Shibuya et al. 2019). The shaded regions associated with the red and blue curves show the 1σ uncertainties of the radial SB profiles. The solid and dashed black lines depict the best-fit Sérsic profiles with $n_s = 1$ and $n_s = 4$, respectively. The gray line denotes a PSF of the imaging data. (b) Redshift evolution of r_e for LAEs (red diamonds; Shibuya et al. 2019) and SFGs+LBGs (small and large blue-filled circles; Shibuya et al. 2015) in the L_{UV} range of $0.12-1 L_{z=3}^*$, where $L_{z=3}^*$ is the characteristic UV luminosity at $z \sim 3$ (Steidel et al. 1999). The red-filled diamonds with and without an open circle represent r_e measured in the rest-frame optical and UV wavelengths, respectively. The red open diamonds are observational estimates of r_e for the luminosity range of $L_{UV} = 0.12-1 L_{z=3}^*$ (see Shibuya et al. 2019). The red region and blue line present the best-fit $(1+z)^\alpha$ functions for the LAEs and SFGs+LBGs, respectively, where α is the power-law index. The gray symbols denote measurements of LAEs with bias-uncontrolled samples that are summarized by Shibuya et al. (2019). Data taken from Shibuya et al. (2019). Abbreviations: LAEs, Lyman- α emitters; LBGs, Lyman break galaxies; PSF, point spread function; SB, surface brightness; SFGs, star-forming galaxies.

This physical picture is explained by the theoretical models of the viewing angle effect in which Ly α photons easily escape in the direction of the disk face-on (Verhamme et al. 2012, Zheng & Wallace 2014). Of course, the morphology of LAEs may not be a simple disk but a disk-like shape with substructures on <1 pkpc scales that are poorly understood. Clumpy structures are identified in various high- z galaxies including some LAEs (Shibuya et al. 2016, Cornachione et al. 2018, Ritondale et al. 2019). Further studies are needed to test the viewing angle effect of the Ly α escape.

5. STELLAR POPULATION

Studying the stellar population of LAEs is important for understanding their physical nature and revealing the relationship between LAEs and other high- z galaxies at similar redshifts. Stellar populations of galaxies are characterized with physical properties such as stellar mass and stellar age and can be investigated from comparisons between their observed spectral energy distributions (SEDs) and those of stellar population synthesis models. Because LAEs are typically faint in the continuum, it is not easy to derive their SEDs on an individual basis, and thus many studies have performed stacking analyses of low spectral-resolution broadband photometry to obtain their typical SEDs with a high signal-to-noise ratio.¹⁰ Early studies have demonstrated

¹⁰There are two major stacking methods, i.e., average and median, and each method has its pros and cons. In general, average stacking can consider all objects in a sample yielding a good representative value, unless the sample includes outliers such as very bright AGNs and/or low- z emission line galaxy contaminants. Median

that the rest-frame UV to optical SEDs of high- z LAEs can be obtained from the combination of deep optical data with NIR images such as those taken with the *Spitzer Space Telescope* (Gawiser et al. 2007, Lai et al. 2008).

Because typical LAEs host star formation, their SEDs of broadband photometry are expected to be characterized with not only stellar emission but also nebular emission from their star-forming regions. In fact, deep NIR spectroscopy results targeting $z \sim 2$ – 3 LAEs have revealed the presence of strong nebular emission lines in the rest-frame optical wavelength such as $H\alpha$ and $[OIII] 5007$ (Guaita et al. 2013; Nakajima et al. 2013). A study of local LAEs indicates that a majority of LAEs have $H\alpha EW_0$ of ~ 50 – $1000 \text{ \AA}$ (Cowie et al. 2010). To include the influence of nebular emission on the determination of stellar population parameters, Ono et al. (2010a) have fitted stellar population synthesis models with and without nebular emission to averaged SEDs of $z \sim 6$ – 7 LAEs by adopting the prescription for nebular emission suggested by Schaerer & de Barros (2009). Ono et al. (2010a) have shown that the best-fit parameters in these two cases are extremely different for the LAEs, mainly because strong nebular lines mimic a substantial Balmer break feature.¹¹ For example, the best-fit stellar mass (age) in the case of considering nebular emission is about an order (two orders) of magnitude smaller than that without nebular emission, which is more serious than the case of the continuum-bright LBGs (Schaerer & de Barros 2009). These results indicate that it is critical to include the effect of nebular emission in stellar population analyses of LAEs.

Subsequent results including the effect of nebular emission have led to a general consensus that typical LAEs are low-mass (stellar mass $\sim 10^{8-9} M_\odot$), young stellar age ($\sim 10 \text{ Myr}$)¹² SFGs with a SFR of ~ 1 – $10 M_\odot \text{ year}^{-1}$ (Figure 9a; Nakajima et al. 2012; see also Hagen et al. 2014), suggesting that typical LAEs are high- z counterparts of local dwarf galaxies, i.e., high- z analogs of local dwarf galaxies. Figure 9b shows compilations of stellar mass and SFR estimates for LAEs at $z \sim 2$ (Hagen et al. 2016) as well as photo- z selected SFGs at similar redshifts (Santini et al. 2017). At fixed stellar masses of $M_* \sim 10^9 M_\odot$, where the completeness for the NB-selected LAEs is high enough for fair comparisons of LAEs and SFGs, LAEs have higher SFRs than the average values of the SFGs, indicating that LAEs have higher specific SFRs than typical SFGs owing to the young stellar ages of LAEs.

6. INTERSTELLAR MEDIUM

In this section, we review four topics of the ISM of LAEs. In the first three topics, we introduce three ISM properties of LAEs: gas kinematics and neutral hydrogen column density, dust extinction, and metallicity. All of these properties are explained by the nature of LAEs that have a $\text{Ly}\alpha$ escape fraction higher than the other galaxy population, which are detailed in the fourth topic. In the fourth topic, we present that LAEs are sources of ionizing photons with distinguished ISM features, i.e., a high ionization state.

6.1. Gas Kinematics and Neutral Hydrogen Column Density

Analyses of emission and absorption lines suggest that LAEs show a strong gas outflow and a low neutral hydrogen column density N_{HI} . The gas kinematics is evaluated from the velocity offset of

stacking is less likely to be affected by such contaminants, although it does not take into account the fluxes of all objects in a sample. Thus, one should check the consistency of the results with these two methods. Note that although these two stacking methods mostly give reasonable results for typical values, no stacking method can identify the large dispersion of properties in a sample if any (Vargas et al. 2014).

¹¹It is confirmed that flux ratios of $[OIII]/H\alpha$ in best-fit SED+nebular emission models are consistent with those directly obtained by spectroscopy (figure 9 of Harikane et al. 2018).

¹²This young stellar age can be found, probably due to intermittent star formation.

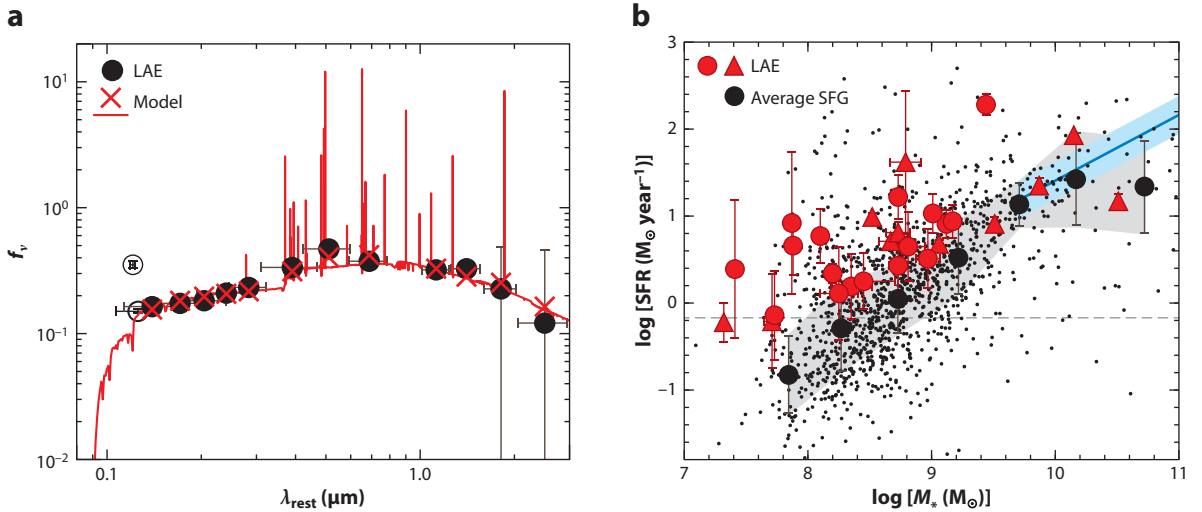


Figure 9

(a) Example of SED fitting results of LAEs. The flux density f_v in arbitrary units is plotted against wavelength in the rest frame. The black-filled (open) circles represent the observed flux densities that are (not) used for SED fitting. The red curve indicates the best-fit SED, and the red crosses correspond to synthesized broadband flux densities that include nebular-emission contributions. Data taken from Nakajima et al. (2012) with permission. (b) SFR as a function of stellar mass. The red circles and triangles are LAEs at $z \sim 2$ that are identified in the narrowband survey and the HETDEX Pilot Survey, respectively, being obtained with model SEDs including nebular emission. The black-dashed horizontal line corresponds to the 5σ detection limit. The black dots are photo- z selected SFGs at similar redshifts, whereas the large black circles correspond to their average values (Santini et al. 2017). The blue solid line and shaded area show the average values and the errors, respectively, for relatively massive SFGs that are regularly studied by deep observations at similar redshifts (Speagle et al. 2014). Data taken from Hagen et al. (2016) with permission. Abbreviations: HETDEX, Hobby-Eberly Telescope Dark Energy Experiment; LAEs, Lyman- α emitters; SED, spectral energy distribution; SFGs, star-forming galaxies; SFR, star-formation rate.

the low-ionization interstellar (LIS) UV metal absorption lines Δv_{IS} with respect to the systemic redshift z_{sys} . Deep NIR spectroscopic observations have found that the LIS UV metal absorption lines are typically blueshifted from z_{sys} by $\sim 200 \text{ km s}^{-1}$ for LAEs (Hashimoto et al. 2013, Shibuya et al. 2014b; **Figure 1a**), suggesting that there exists a notable gas outflow with an outflow velocity of $V_{\text{exp}} \sim 200 \text{ km s}^{-1}$ in LAEs that is similar to LBGs, the majority of which are SFGs with weak-to-no Ly α emission (Steidel et al. 2010). Another important tool characterizing the gas kinematics is a Ly α emission line. Observations of Ly α emission find various shapes of Ly α profiles, a Ly α emission velocity offset (redshift) from the systemic velocity and double Ly α emission line peaks. Several studies have measured the Ly α velocity offset $\Delta v_{\text{Ly}\alpha}$ for LAEs (McLinden et al. 2011, Guaita et al. 2013) and have identified an anticorrelation between Ly α EW_0 and $\Delta v_{\text{Ly}\alpha}$ (Hashimoto et al. 2013, Erb et al. 2014, Shibuya et al. 2014b). As shown in **Figure 10a**, LAEs have a Ly α velocity offset of $\Delta v_{\text{Ly}\alpha} \sim 200 \text{ km s}^{-1}$ (see a typical spectrum in **Figure 1e**), which is systematically smaller than $\Delta v_{\text{Ly}\alpha} \sim 400 \text{ km s}^{-1}$ of LBGs (Steidel et al. 2010; see a typical spectrum in **Figure 1c**). According to the observational results, LAEs have, on average, a Ly α velocity offset comparable with the outflow velocity, $\Delta v_{\text{Ly}\alpha} \sim V_{\text{exp}} \sim 200 \text{ km s}^{-1}$, whereas LBGs show $\Delta v_{\text{Ly}\alpha} \sim 2 V_{\text{exp}} \sim 400 \text{ km s}^{-1}$.

If the ES model (Section 2.2) is applicable to LAEs and LBGs, various profiles of the observed Ly α emission lines including the Ly α velocity offsets and double peaks can be beautifully explained. The ES models best-fit to the observational data are presented with red lines in **Figure 1c,e**. Furthermore, the Ly α EW_0 - $\Delta v_{\text{Ly}\alpha}$ anticorrelation can be interpreted by the difference in N_{HI} between

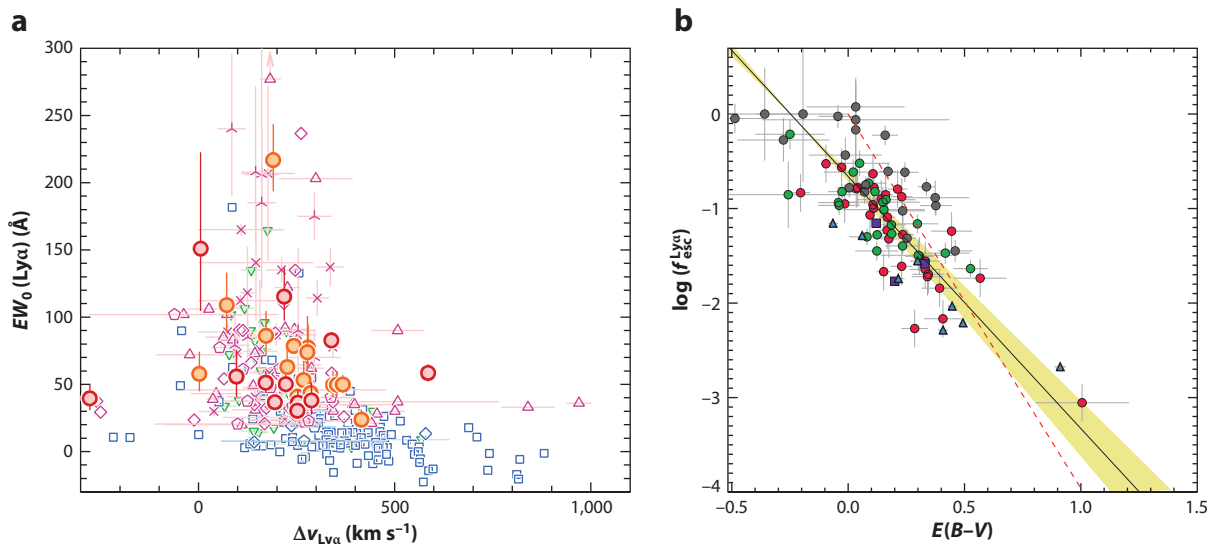


Figure 10

(a) Ly α EW_0 as a function of Ly α velocity offset $\Delta v_{\text{Ly}\alpha}$. The data points are the compilation of the literature data (*red, orange, and magenta*: LAEs at $z \sim 2-3$; *blue*: Lyman break galaxies at $z \sim 2-3$; *green*: green pea galaxies at $z \sim 0.3$). (b) Ly α escape fraction as a function of color excess, $E(B-V)$. The data points represent measurements of star-forming galaxies at $z \sim 0-0.3$ taken from the literature. The differences of the data-point colors indicate differences of the references. The black line and the yellow shading present the best-fit linear function and its 1σ fitting uncertainty, respectively. The red-dashed line denotes the Milky Way extinction law for a foreground-screen case that ignores Ly α resonant scattering. Note that a similar relation is also found for LAEs at $z = 2-4$ (Blanc et al. 2011). Panel *a* adapted from Nakajima et al. (2018) with permission, and panel *b* from Atek et al. (2014) with permission. Abbreviations: EW_0 , rest-frame equivalent width; LAE, Lyman- α emitter.

LAEs and LBGs. In the ES model, the relation of $\Delta v_{\text{Ly}\alpha} \sim V_{\text{exp}}$ is obtained; if the backscattered Ly α emission is weak due to the small H I column density of the shell, $N_{\text{HI}} \lesssim 10^{20} \text{ cm}^{-2}$. In contrast, $\Delta v_{\text{Ly}\alpha} \sim 2 V_{\text{exp}}$ is reproduced if the amount of Ly α back scattering is large in the case of a high H I column density in the shell, $N_{\text{HI}} \gtrsim 10^{20} \text{ cm}^{-2}$. The interpretation based on the ES model suggests that LAEs have an H I column density of the shell lower than those of LBGs, which are quantitatively concluded by ES model fitting results (Hashimoto et al. 2015). This difference in N_{HI} would produce the Ly α EW_0 - $\Delta v_{\text{Ly}\alpha}$ anticorrelation.

The low H I column density for LAEs is also explained with measurements of the covering fraction, f_c , of LIS gas estimated from the depth of LIS metal absorption lines. Medium-high-resolution spectroscopy finds a possible anticorrelation between Ly α EW_0 and f_c (Jones et al. 2013, Shibuya et al. 2014b). Because f_c positively correlates with the H I gas density on average, the Ly α EW_0 - f_c anticorrelation supports the idea that a large EW_0 (i.e., a large Ly α escape) is given by the low N_{HI} gas clouds.

6.2. Dust Extinction

LAEs are known as dust poor galaxies. The stellar and nebular extinction values are estimated to be $E(B-V) \sim 0-0.2$ for the majority of LAEs by stellar-population synthesis model fitting and Balmer decrement analyses, respectively (Ono et al. 2010b, Kojima et al. 2017). The dust-poor nature of LAEs is confirmed with measurements of the UV-continuum spectral slope β that is defined by the power-law approximation of the UV-continuum flux, $f_\lambda \propto \lambda^\beta$. LAEs have typically

a blue UV-continuum spectrum of $\beta \sim -2$ that is systematically smaller than those of LBGs in the same UV magnitude range (Stark et al. 2010), supporting the fact that LAEs are dust poor.

The dust extinction law of a galaxy can be evaluated by combining β and the *IRX* ratio,

$$IRX = \frac{L_{IR}}{L_{UV}}, \quad 10.$$

where L_{IR} and L_{UV} are the total IR and UV luminosities, respectively. Recent studies using *Spitzer* and the *Herschel Space Observatory* have revealed that LAEs at $z \sim 2$ have typically low *IRX* values at a given β value similar to that of the Small Magellanic Cloud (SMC) or even below those of Calzetti's local starbursts (Wardlow et al. 2014, Kusakabe et al. 2015). These low *IRX* values are consistent with results that LAEs at $z > 5$ have faint far-infrared (FIR) continua at the observed-frame 1-mm band found by Atacama Large Millimeter/submillimeter Array (ALMA) observations (Knudsen et al. 2016). These *IRX* and FIR studies also confirm that LAEs have low dust extinction.

6.3. Metallicity

LAEs have a gas-phase metallicity lower than that of SFGs with a low Ly α EW based on deep spectroscopy and two kinds of metallicity estimates, strong line and direct electron temperature T_e methods. The strong line method using, e.g., H α , H β , and [OIII] 5007,4959 is widely applied to LAEs at $z \sim 2-3$. Observational studies have estimated the typical gas-phase metallicity to be $Z \sim 0.1-0.5 Z_\odot$ for LAEs at $z \sim 2-3$ with the N2 and R23 indices (Finkelstein et al. 2011, Nakajima et al. 2012). These metallicity measurements are comparable with or lower than those of the galaxy mass-metallicity relation at $z \sim 2-3$ (Finkelstein et al. 2011) and consistent with the SFR-mass metallicity relation (Nakajima et al. 2012). Similarly, low metal abundances, $Z \sim 0.1-0.3 Z_\odot$, are obtained for $z \sim 2-4$ LAEs by the T_e methods with the successful detections of faint T_e -sensitive emission lines, [OIII] 4363 (Trainor et al. 2016) and OIII] 1661,1666 (Kojima et al. 2017). These studies of strong line and direct T_e methods indicate that the gas-phase metallicity of typical LAEs falls in a range of $Z \sim 0.1-0.5 Z_\odot$ that is similar to or slightly lower than the typical metallicity of LBGs with the same UV-continuum luminosity (Steidel et al. 2014).

LAEs with a large Ly α EW_0 are thought to be candidates for metal-free, i.e., Population III (Pop III) galaxies. A large Ly α EW_0 of $\gtrsim 240 \text{ \AA}$ cannot be reproduced by recombination processes in a normal SFG, i.e., Population II (Pop II) galaxy, with the solar metallicity and a Salpeter initial mass function (IMF; Malhotra & Rhoads 2002). **Figure 11a** presents theoretical predictions for Ly α EW_0 of young galaxies including Pop III galaxies. The large Ly α EW_0 can be accomplished by a young extremely metal-poor galaxy with a top-heavy IMF that efficiently produces ionizing photons for strong Ly α via recombination (Schaerer 2003). Large-area imaging and deep spectroscopic observations have identified a number of large Ly α EW sources (Hashimoto et al. 2017, Shibuya et al. 2018b). A strong HeII1640 emission line is another useful tool to distinguish Pop III galaxies from Pop II galaxies, because the production of HeII line requires metal-poor massive stars emitting high-energy ($> 54 \text{ eV}$) photons that ionize He $^+$. **Figure 11b** shows HeII1640 EW_0 as a function of stellar age for models of young galaxies including Pop III galaxies and indicates that a moderately strong HeII1640 ($EW_0 \gtrsim 5 \text{ \AA}$) line can be observed in a young extremely metal-poor galaxy with a top-heavy IMF. Although HeII1640 emission can be also found in AGNs and SFGs with a notable amount of Wolf-Rayet stars, such objects, except type 2 AGNs and high-mass X-ray binaries, can be removed by the presence of their broad line ($\gtrsim 1000 \text{ \AA}$) component of HeII emission. There are several reports of detecting such strong and narrow HeII1640 emission in large Ly α EW sources (Prescott et al. 2009, Sobral et al. 2015). However, subsequent spectroscopy and multiwavelength studies suggest that these objects are not Pop III galaxies due to the detections of metal lines or the lack of evidence on the strong HeII1640 emission (Prescott et al. 2015, Bowler

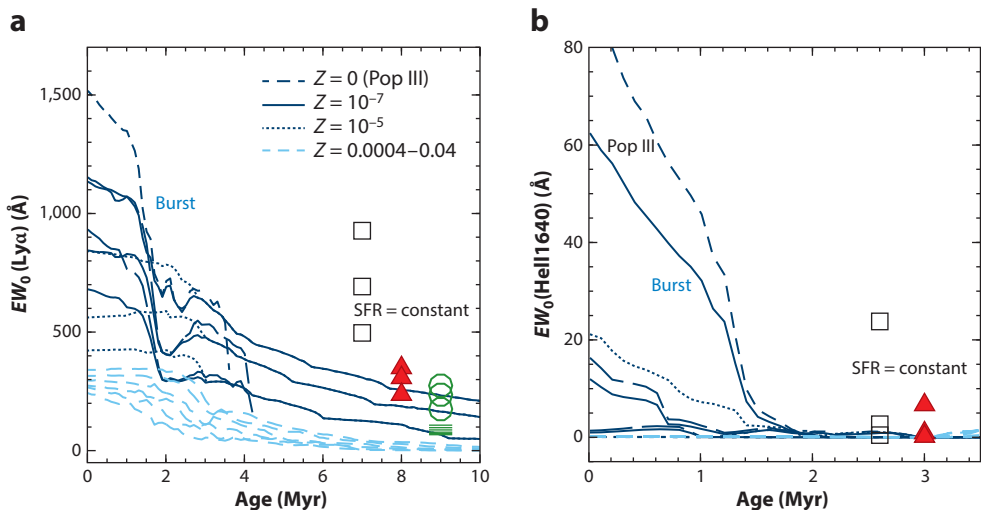


Figure 11

(a) Ly α EW_0 as a function of stellar age for models of stellar evolution and photoionization (Schaerer 2003). The three blue dashed lines indicate Pop III ($Z = 0$) instantaneous starburst models for a Salpeter IMF with mass ranges of 50–500 $M_\odot$ (top), 1–500 $M_\odot$ (middle), and 1–100 $M_\odot$ (bottom). The blue solid (dotted) lines are the same as the blue dashed lines but for a metallicity of $Z = 10^{-7}$ (10^{-5}). The cyan dashed lines represent instantaneous starbursts of Salpeter IMF 1–100 $M_\odot$ models with $Z = 0.0004, 0.001, 0.004, 0.008, 0.020$, and 0.040 from top to bottom. The three squares denote constant SFR models of the 50–500- $M_\odot$ Salpeter IMF with metallicities of $Z = 0, 10^{-7}$, and 10^{-5} from top to bottom, where the stellar ages are arbitrary. The red triangles (green circles) are the same as the squares but for the mass range of 1–500 $M_\odot$ (1–100 $M_\odot$). The green lines are the same as the green circles but for metallicities of $Z = 0.0004, 0.001, 0.004, 0.008, 0.020$, and 0.040 from top to bottom. (b) Same as panel a but for HII1640 EW_0 . Figure adapted from Schaerer (2003) with permission. Abbreviations: EW_0 , rest-frame equivalent width; IMF, initial mass function; Pop III, Population III; SFR, star-formation rate.

et al. 2017, Shibuya et al. 2018a; see section 4.3 of Ouchi 2019 for a summary). So far, no promising candidates of Pop III galaxies have remained.

6.4. Ly α Escape Fraction

As described above, LAEs have the ISM with N_{HI} , $E(B-V)$, and Z lower than those of other types of galaxies with a low Ly α EW at a similar redshift. These characteristics of the ISM are plausibly related to the high Ly α escape fraction ($f_{\text{esc}}^{\text{Ly}\alpha}$; Section 3) of LAEs. The low N_{HI} and $E(B-V)$ allow Ly α photons to easily escape from the ISM of LAEs due to a small number of Ly α resonant scattering and absorption events. Here, the low Z is necessary for the low $E(B-V)$.

The connection between $f_{\text{esc}}^{\text{Ly}\alpha}$ and ISM properties is understood on the individual and statistical bases. The Ly α escape fraction has been estimated individually for LAEs at $z \sim 0-4$ (Blanc et al. 2011, Atek et al. 2014). **Figure 10b** shows that $f_{\text{esc}}^{\text{Ly}\alpha}$ anticorrelates with $E(B-V)$, indicating that Ly α photons are heavily absorbed in a dusty galaxy. Interestingly, as discussed in Section 3 (and shown in **Figure 7**), the cosmic average $f_{\text{esc}}^{\text{Ly}\alpha}$ values increase monotonically from $z \sim 0$ to $z \sim 6$ by two orders of magnitude (Hayes et al. 2010, Konno et al. 2016). The strong (two orders of magnitude) increase in $f_{\text{esc}}^{\text{Ly}\alpha}$ is explained by neither (1) IGM absorption, (2) stellar population, (3) outflow velocity, (4) clumpy ISM, nor (5) dust extinction of a simple screen dust model (Konno et al. 2016).

Historically, it was often discussed that the large $f_{\text{esc}}^{\text{Ly}\alpha}$ found in SFGs at high redshift was explained by the scenario of selective dust attenuation for Ly α and UV continuum in the clumpy ISM (a.k.a. Neufeld's effect; Neufeld 1991). In the clumpy ISM, Ly α photons are resonantly scattered on the surface of clumpy gas clouds with a negligible dust absorption (Neufeld 1991, his figure 1). By the resonant scattering, the total number of Ly α photons is conserved. By contrast, UV-continuum photons penetrate multiple clumpy gas clouds in the foreground whose centers have dusty molecular gas, being heavily absorbed by the dust. The difference of Ly α and UV-continuum extinction produces a high Ly α EW_0 . The validity of Neufeld's effect had not been tested in realistic ISM conditions but only in a simple case of the static and very clumpy and/or dusty ISM since the publication of Neufeld (1991). Recent radiative transfer simulations have conducted extensive calculations with various physical parameters and found that Neufeld's effect exists but emerges only under special physical conditions: low outflow velocity, very high extinction, and extremely clumpy gas distribution (most gas is locked up in clumps). Because these three characteristics do not agree with the observed properties of LAEs (Laursen et al. 2013, Duval et al. 2014), the clumpy ISM scenario is probably not the reason for the high $f_{\text{esc}}^{\text{Ly}\alpha}$ values at high redshift.

Although the physical origin of the strong increase in $f_{\text{esc}}^{\text{Ly}\alpha}$ has not been definitively concluded, the rise in $f_{\text{esc}}^{\text{Ly}\alpha}$ is probably caused by another scenario: (6) the decrease of H I column density toward high z that reduces the resonant scattering and, thus, dust attenuation of Ly α . This scenario can explain the increase of $f_{\text{esc}}^{\text{Ly}\alpha}$ by two orders of magnitude (Konno et al. 2016).

6.5. Ionization State

Recent optical and NIR spectroscopy find that LAEs have ionizing states significantly different from those of other galaxy populations. **Figure 12a** shows that the O32 ratio,¹³ the line flux ratio of [OIII] 5007 to [OII] 3727, of $z \sim 2-3$ LAEs is $\sim 10-100$ times higher than those of local galaxies and even higher than those of LBGs at similar redshifts (Nakajima & Ouchi 2014). Based on comparisons with photoionization models in the O32 versus R23¹⁴ diagram, the high O32 ratios of LAEs indicate high ionization parameters, $q_{\text{ion}} = 1-9 \times 10^8 \text{ cm s}^{-1}$. Detections of high ionization lines of CIII] 1907, 1909 and CIV 1548 also indicate that the ionization state of LAEs is very high (Stark et al. 2014). However, recent ALMA studies have suggested that the [CII] 158 μm emission of LAEs are systematically fainter than the local SFR- $L_{[\text{CII}]}$ relation (Ouchi et al. 2013, Harikane et al. 2018; cf. Carniani et al. 2018). Although a faint [CII] 158 μm luminosity for a given SFR could be explained by the presence of AGNs or the collisional deexcitation of C⁺, LAEs show neither AGN activity nor gas density high enough for the collisional deexcitation.

The two properties of the high q_{ion} and weak [CII] 158- μm emission may be consistently explained by the ISM of density-bounded nebulae (Nakajima & Ouchi 2014). **Figure 12b** shows a conceptual diagram of density-bounded and ionization-bounded nebulae. In the density-bounded (ionization-bounded) nebula, the size of ionized regions is determined by the amount of gas around ionizing sources (the number of ionizing photons). In contrast with the ionization-bounded nebula, the density-bounded nebula has a small outer shell of ionized gas emitting low-ionization lines including [OII] 3727, whereas the density-bounded nebula has a well-developed inner core of highly ionized gas producing high-ionization lines (e.g., CIII] 1907, 1909, [OIII] 5007, 4959, and [OIII] 88 μm). In this case, a [CII] 158 μm -emitting photodissociation region (PDR) does not exist. More realistically, a nonzero $\sim 10\%$ covering fraction of PDR can

¹³In this review, we use the definition, $\text{O32} \equiv f([\text{OIII}]5007)/f([\text{OII}]3727)$. Note that some studies use $\text{O32} = f([\text{OIII}]5007, 4959)/f([\text{OII}]3727)$ and that the ratio of $f([\text{OIII}]5007)$ to $f([\text{OIII}]4959)$ is 3, which is a constant value determined by atomic physics (Storey & Zeippen 2000).

¹⁴ $\text{R23} \equiv \{f([\text{OIII}]5007, 4959) + f([\text{OII}]3727)\}/f(\text{H}\beta)$.

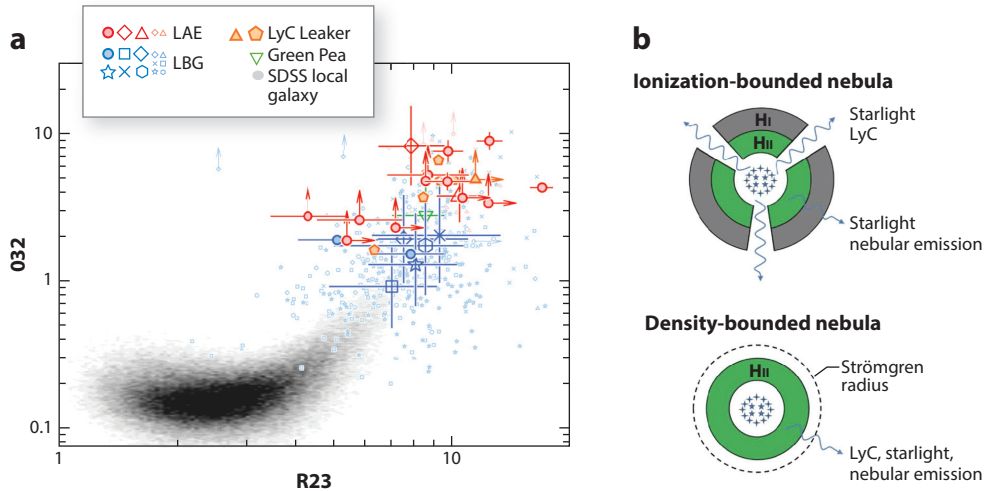


Figure 12

(a) Diagram of O32 versus R23 indices for high- z and local galaxies. The red and blue symbols represent LAEs and LBGs at $z \sim 2-4$, respectively. The orange data points indicate Lyman-continuum leakers. The green triangle denotes a green pea galaxy. The gray shading shows SDSS local galaxies. The large open symbols and the cross indicate average measurements. Data taken from Nakajima & Ouchi (2014) and Nakajima et al. (2016) with permission. (b) Conceptual figure for ionization-bounded (*top*) and density-bounded (*bottom*) nebulae. Panel *b* adapted from Zackrisson et al. (2013) with permission. Abbreviations: LAE, Lyman- α emitter; LBG, Lyman break galaxy; LyC, Lyman continuum; SDSS, Sloan Digital Sky Survey.

explain both small [C II] 158 μm -to-SFR and large [O III] 88 μm -to-SFR ratios found in high- z SFGs, a majority of which are LAEs (Harikane et al. 2020). Furthermore, the density-bounded nebula would help ionizing photons escape from the ISM of LAEs, contributing to the cosmic reionization (Jaskot & Oey 2014, Nakajima & Ouchi 2014; see Section 9).

There are many open questions about the ISM of LAEs. Although the density-bounded nebula is an interesting scenario, no direct evidence of density-bounded nebulae has been obtained. Another problem of the ISM is that some LAEs have an extremely low IRX value for a given UV slope β even below the SMC's dust extinction curve (Capak et al. 2015). It is unclear how such a low IRX value is reproduced, because realistic dust sizes cannot make the flat IRX - β relation. Besides these detailed physical questions, one should push high Ly α EW LAE searches (see the discussion of metallicity in Section 6.3) for a promising candidate of a Pop III galaxy whose identification is key for understanding the first stage of galaxy formation.

7. CIRCUMGALACTIC MEDIUM AND THE LARGE-SCALE STRUCTURES TRACED BY DIFFUSE LYMAN- α

This section presents the diffuse and spatially extended Ly α emission around SFGs from a small scale to a large scale that are known as Ly α halos (LAHs), Ly α blobs (LABs), enormous Ly α nebulae (ELANe), and the large-scale Ly α emission in the IGM. The extended Ly α emission has a large diversity in size, ranging from ~ 1 pkpc to $\gtrsim 1,000$ pkpc, which is summarized in **Figure 13**. Note that LAHs, LABs, and ELANe are not clearly defined by physical quantities but roughly classified by the Ly α luminosity and size. Although the physical origins of extended Ly α emission are poorly understood, there are six scenarios for extended Ly α emission: (1) resonant scattering

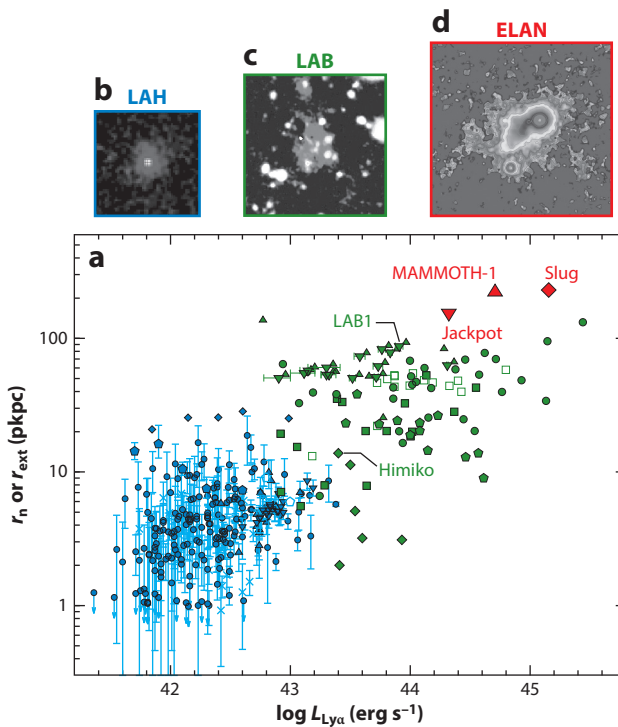


Figure 13

(a) Ly α scale-length r_n or extent r_{ext} of extended Ly α emission as a function of Ly α luminosity. The blue, green, and red symbols denote LAHs, LABs, and ELANs, respectively. Data taken from the literature (Matsuda et al. 2011; Steidel et al. 2011; Cantalupo et al. 2014; Momose et al. 2014, 2016; Hennawi et al. 2015; Borisova et al. 2016; Cai et al. 2016; Wisotzki et al. 2016; Leclercq et al. 2017; Xue et al. 2017; Zhang et al. 2019). The definitions of r_{ext} and Ly α luminosity depend on various studies in the literature. Here, the r_{ext} values are converted from the end-to-end Ly α extent (Borisova et al. 2016) multiplied by a factor of 0.5 to mitigate the difference in measurement techniques. Panels *b–d* show sample snapshots of an LAH, LAB, and ELAN and are adapted with permission from Leclercq et al. (2017), Matsuda et al. (2011), and Cantalupo et al. (2014), respectively. Abbreviations: ELAN, enormous Lyman- α nebula; LAB, Lyman- α blob; LAH, Lyman- α halo.

of Ly α emission from central star-forming regions and/or AGNs, (2) photoionization and recombination in unresolved dwarf satellite galaxies, (3) outflowing gas, (4) infalling gas, (5) fluorescence, and (6) galaxy mergers. In the following paragraphs, we review properties and possible physical origins of the extended Ly α emission.

7.1. Ly α Halos

LAHs are the 1–10-pkpc-scale extended Ly α emission found around SFGs at $z \sim 2$ –7 (**Figure 14a–d**). Although the Ly α luminosity and size of LAHs depend on physical properties of host SFGs, most high- z SFGs ubiquitously have LAHs. The diffuse Ly α emission of LAHs have been identified by image stacking analyses on the statistical basis (Hayashino et al. 2004, Steidel et al. 2011, Matsuda et al. 2012) and deep spectroscopic observations on an individual basis (Wisotzki et al. 2016, Leclercq et al. 2017). The Ly α radial SB profile is approximated with an exponential function,

$$S(r) = C_n \exp(-r/r_n), \quad 11.$$

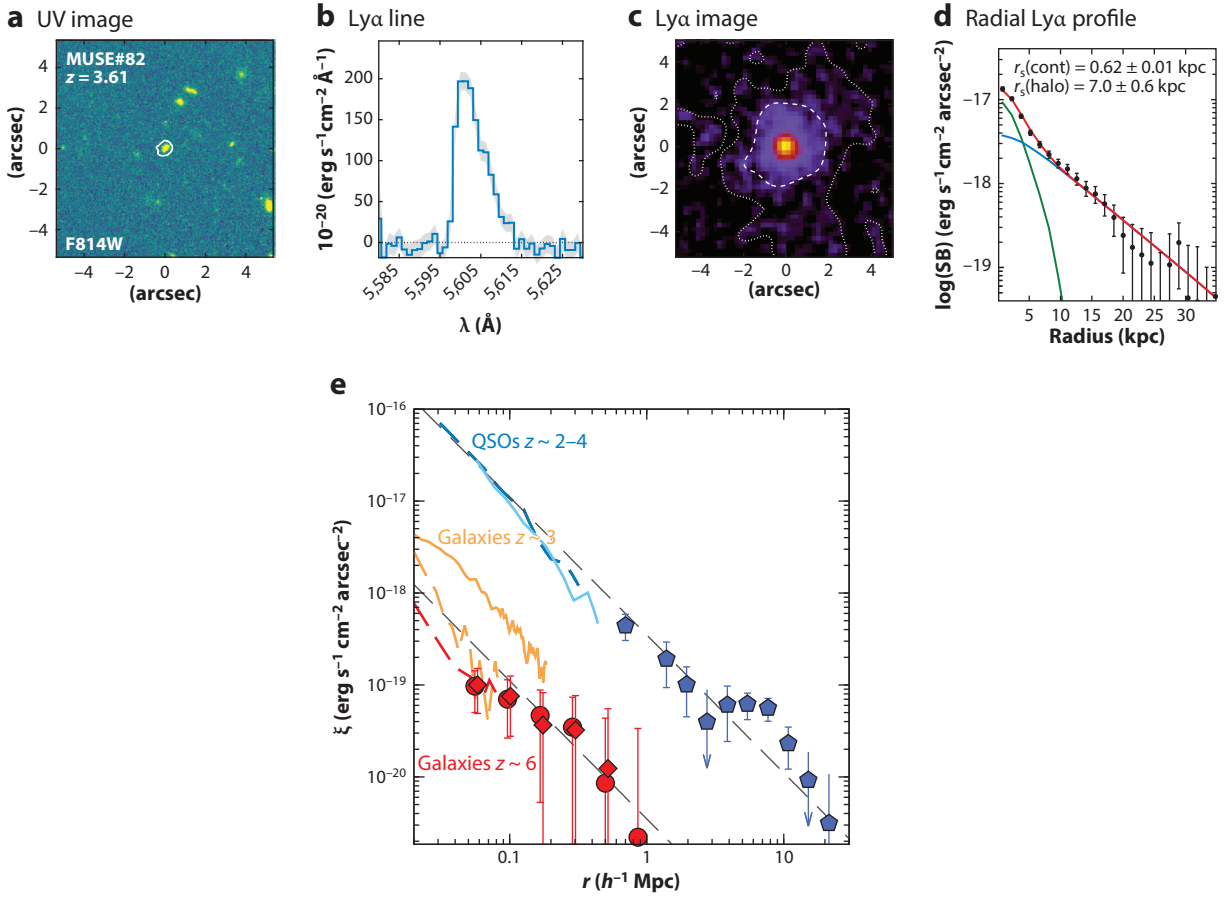


Figure 14

(a–d) Images, spectrum, and radial SB profiles of a galaxy with an LAH at $z = 3.61$. (a) UV-continuum image. (b) $\text{Ly}\alpha$ spectrum. (c) $\text{Ly}\alpha$ image. (d) Radial $\text{Ly}\alpha$ SB profile. In panel d, the black circles with the error bars present the observed $\text{Ly}\alpha$ SB profile, whereas the (red) green and blue curves indicate the best-fit $\text{Ly}\alpha$ SB profiles of (total) core and LAH components, respectively. (e) Cross-correlation functions of $\text{Ly}\alpha$ emission with SDSS BOSS quasars at $z \sim 2\text{--}4$ (blue pentagons; Croft et al. 2018) and with $\text{Ly}\alpha$ emitters (red circles: $z = 5.7$; red diamonds: $z = 6.6$; Kakuma et al. 2019). The curves represent radial $\text{Ly}\alpha$ SB profiles of an enormous $\text{Ly}\alpha$ nebula (cyan solid line: Cantalupo et al. 2014), radio-quiet quasars (blue dashed line: Borisova et al. 2016), and galaxies at $z \sim 3$ (orange solid line: Steidel et al. 2011; orange dashed line: Leclercq et al. 2017) and at $z \sim 6$ (red dashed line: Leclercq et al. 2017). The two black dashed lines are power-law functions of $\xi \propto r^{-1.5}$ that are shown for reference. Panels a–d adapted from Leclercq et al. (2017) with permission. Abbreviations: BOSS, Baryon Oscillation Spectroscopic Survey; LAH, Lyman- α halo; QSO, quasi-stellar object; SB, surface brightness; SDSS, Sloan Digital Sky Survey.

where $S(r)$, r , and C_n are the $\text{Ly}\alpha$ SB, the radius from the galaxy center, and the normalization factor, respectively.¹⁵ The parameter of r_n is the LAH scale length, quantifying the extent of LAHs. Several studies find that r_n correlates with physical properties of LAEs, providing useful hints for physical origins of LAHs. In SFGs, the LAH scale length decreases with increasing $\text{Ly}\alpha$ luminosity

¹⁵Although the exponential profile is an empirical model, a $\text{Ly}\alpha$ radial SB profile similar to the exponential profile can be derived on the basis of a physical picture of $\text{Ly}\alpha$ resonant scattering in a gaseous halo with a simple power-law distribution of H I gas covering fraction, $f_c \propto r^{-\gamma}$, where γ is a power-law index (Steidel et al. 2011).

of the central part of the SFGs at $r < 8$ pkpc ($L_{\text{cent, Ly}\alpha}$; Momose et al. 2016), indicating that a large LAH is found in $L_{\text{cent, Ly}\alpha}$ -faint galaxies. (In other words, LAHs are small in LAEs whose central < 8 -pkpc part of $\text{Ly}\alpha$ luminosity is bright.) The $L_{\text{cent, Ly}\alpha}$ -faint galaxies may be explained by strong $\text{Ly}\alpha$ resonant scattering (scenario 1) in the thick H I clouds near the galaxy center, which eventually produces a largely extended LAH. However, radiative transfer and hydrodynamical simulations suggest that the $\text{Ly}\alpha$ resonant scattering effect (scenario 1) cannot reproduce the largely extended $\text{Ly}\alpha$ SB profiles of LAHs, whereas the cooling radiation (scenario 4) may be needed to explain $\text{Ly}\alpha$ emission at the outer part of LAHs (Lake et al. 2015). In contrast with the $\text{Ly}\alpha$ emission, UV-continuum emission is not spatially extended in observational data (Momose et al. 2014, Leclercq et al. 2017). This compact UV-continuum emission would rule out significant contribution from unresolved dwarf satellite galaxies (scenario 2). Recently, stacked ALMA data reveal that SFGs at $z \sim 5\text{--}7$ have a 10-pkpc-scale extended carbon [C II] 158 μm emission whose SB profile shape is very similar to that of the $\text{Ly}\alpha$ emission (Fujimoto et al. 2019). The existence of the extended emission of [C II] similar to $\text{Ly}\alpha$ might suggest that LAHs originated from the neutral gas expelled by outflows that are carbon enriched (scenario 3). Deep observations identify that the r_n value is almost constant over the redshift range of $z \sim 2\text{--}6$, perhaps suggesting no significant redshift evolution of physical properties of LAHs at $z < 6$ (Momose et al. 2014, Leclercq et al. 2017). There is a hint of an increase of r_n toward the reionization epoch of $z = 6.6$, but the increase is found only at the 1σ level (Momose et al. 2014).

7.2. $\text{Ly}\alpha$ Blobs

LABs are spatially extended $\text{Ly}\alpha$ nebulae with a physical scale of $\sim 10\text{--}100$ pkpc and a $\text{Ly}\alpha$ luminosity of $\sim 10^{43}$ erg s $^{-1}$ (Steidel et al. 2000, Matsuda et al. 2004, Shibuya et al. 2018b). Such extended $\text{Ly}\alpha$ emission has been found around various types of sources: e.g., SFGs (Steidel et al. 2000, Ouchi et al. 2009), radio-loud galaxies (McCarthy et al. 1987, van Ojik et al. 1997), and radio-quiet QSOs (Borisova et al. 2016).

Physical properties of LABs have been investigated by multiwavelength observational and theoretical studies. Deep X-ray observations find that a majority of LABs show no clear AGN activity, whereas $\sim 20\%$ of LABs host an AGN (Basu-Zych & Scharf 2004, Geach et al. 2009). Similarly, $\sim 30\%$ of LABs are detected at radio wavelengths (Ao et al. 2017). These X-ray and radio results would indicate that AGNs contribute to the extended $\text{Ly}\alpha$ emission in some LABs via $\text{Ly}\alpha$ resonant scattering and/or fluorescence (scenarios 1 and 5). A tangential polarization signal up to 20% is detected in the $\text{Ly}\alpha$ emission around an LAB (Hayes et al. 2011), which might suggest that LABs are produced by $\text{Ly}\alpha$ resonant scattering (scenario 1; Dijkstra & Loeb 2008; cf. Trebitsch et al. 2016). According to theoretical studies (Fardal et al. 2001, Dijkstra & Loeb 2009, Goerdt et al. 2010), the extended $\text{Ly}\alpha$ emission of LABs can be reproduced by the cooling radiation (scenario 4). However, LABs have a moderately large $\text{Ly}\alpha$ velocity offset from the systemic velocity, several hundred kilometers per second (Yang et al. 2014), that may be evidence for strong outflow (scenario 3). HST observations reveal that some high- z LABs have multiple stellar components (Ouchi et al. 2013, Sobral et al. 2015), which suggests that starbursts caused by galaxy mergers (scenario 6) create the extended $\text{Ly}\alpha$ emission. The galaxy merger scenario might be supported by the fact that LABs reside in galaxy overdense regions (Matsuda et al. 2004, Yang et al. 2010, Kikuta et al. 2019, Harikane et al. 2019).

7.3. Enormous $\text{Ly}\alpha$ Nebulae

ELANes are extended $\text{Ly}\alpha$ emission harboring QSOs (Cantalupo et al. 2014, Hennawi et al. 2015, Cai et al. 2016). The $\text{Ly}\alpha$ emission of ELANes extends to a scale of several hundred pkpc

(>400 pkpc for some ELANe), which is larger than the virial radius of a dark matter halo that hosts the QSOs. The existence of energetic QSOs suggests that ELANe are fluorescently illuminated (scenario 5). However, very high density clumps are needed to produce the observed high Ly α SB of $\sim 10^{-18}$ – 10^{-16} erg s $^{-1}$ cm $^{-2}$ arcsec $^{-2}$ (Cantalupo et al. 2014). Alternatively, the Ly α resonant scattering may be also important in ELANe (scenario 1). Distinguishing the two scenarios of (1) and (5), one should observe nonresonant hydrogen recombination lines such as H α in ELANe as demonstrated by Leibler et al. (2018).

NB imaging observations and the Keck/Palomar Cosmic Web Imagers (KCWI/PCWI) spectroscopy find that ELANe tend to reside in galaxy overdense regions (Cai et al. 2017, 2018). The environments of ELANe indicate that ELANe would exist in progenitors of massive galaxy clusters.

7.4. The Large-Scale Ly α Emission

The large-scale Ly α emission extending to ~ 1 – 15 cMpc is identified by the cross-correlation Ly α intensity mapping technique (**Figure 14e**). The cross-correlation Ly α intensity mapping technique obtains the spatial cross correlation between objects with known redshifts (e.g., galaxies and QSOs) and faint Ly α emission below a detection limit of imaging or spectroscopic data and to detect the faint Ly α emission, where the spatial cross-correlation analysis systematically removes signals unrelated to the objects. Croft et al. (2018) report the detection of very faint Ly α emission around QSOs at the large scale of ~ 1 – 15 cMpc by Ly α intensity mapping of cross correlation between Sloan Digital Sky Survey (SDSS) Baryon Oscillation Spectroscopic Survey (BOSS) QSOs at $z \sim 2$ – 3.5 and Ly α emission in a large number of SDSS spectra (with no Ly α detections on an individual basis). The radial SB profile of this large-scale Ly α emission is smoothly connected to the ELANe at $r \sim 1$ cMpc (**Figure 14e**). The connection with ELANe indicates that the large-scale Ly α emission may be produced by the fluorescent radiation from QSOs (scenario 5). In addition to QSOs, Kakuma et al. (2019) conduct Ly α intensity mapping of cross correlation between LAEs at $z \sim 6$ and Ly α emission in NB image pixels and identify extremely faint Ly α emission around LAEs at a spatial scale beyond a virial radius of the LAE–dark matter halo host (~ 0.15 cMpc) up to ~ 1 cMpc. These cross-correlation signals of LAEs are consistent with the extrapolation of Ly α radial SB profiles of LAHs (Leclercq et al. 2017; **Figure 14e**). Comparisons with numerical simulations suggest that the large-scale Ly α emission is not fully explained by the combination of resonant scattering (scenario 1) and unresolved dwarf satellite galaxies (scenario 2) but rather is contributed by Ly α emission created by other mechanisms that possibly include cold accretion.

Due to diverse physical properties, it is still under debate how the extended Ly α emission is produced in LAHs, LABs, and ELANe. It is also unclear whether the possible increase in r_n to the EoR truly exists (see the topic of LAHs in Section 7.1), which is thought to be an indicator of the IGM neutral hydrogen fraction increase (Jeeson-Daniel et al. 2012).

8. CLUSTERING OF LYMAN- α EMITTERS

Distant cosmic structures including primeval large-scale structures and progenitors of galaxy clusters (a.k.a. protoclusters) are investigated with LAEs. LAEs are advantageous for mapping out the spatial distribution of high- z galaxies efficiently by NB imaging and spectroscopic surveys (Ouchi et al. 2005, Yamada et al. 2012). **Figure 15** is the map of LAEs at $z \sim 6$ over the scale of 1.5 deg 2 corresponding to ~ 200 cMpc, showing primeval galaxy clusters, filaments, and voids.

The systematic large-scale structure survey is being conducted by HETDEX (Section 3). HETDEX is a cosmology survey investigating cosmic structures at $z \sim 2$ – 3 , measuring the

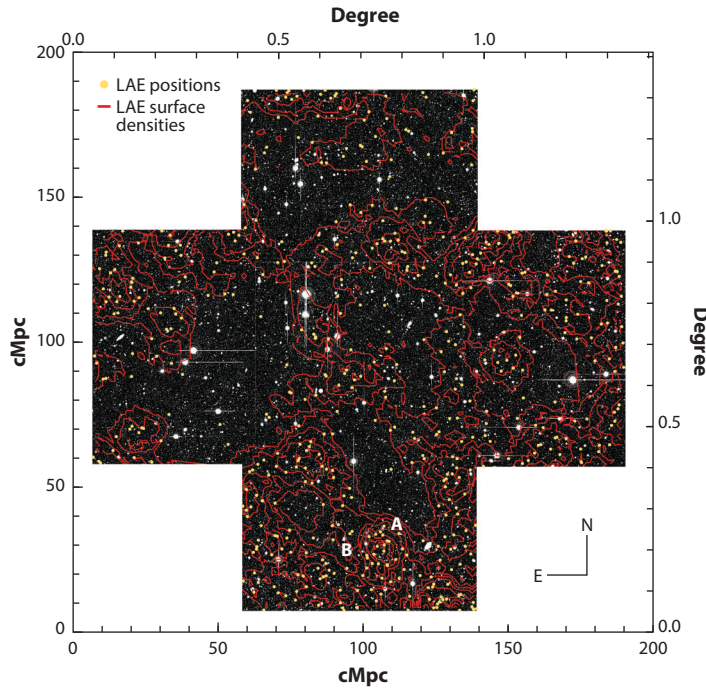


Figure 15

Map of LAEs at $z = 5.7 \pm 0.05$ over the scale of 1.5 deg^2 that corresponds to $\sim 200 \text{ cMpc}$ (Ouchi et al. 2005). The yellow dots represent positions of the LAEs on the survey area of a black cross-hair region that is indicated with the color composite sky image. The red contours denote the surface density of the LAEs. The positions of the LAE overdensity regions are marked with the characters, A and B. Figure adapted from Ouchi et al. (2005) with permission. Abbreviations: cMpc, comoving megaparsec; LAE, Lyman- α emitter.

baryon acoustic oscillation on the scale of $\sim 100 \text{ cMpc}$ for understanding the equation of state of dark energy.

On the scale of galaxy clusters, LAEs are tracers of the galaxy distribution, which is the same concept as that used by the HETDEX cosmology survey. The overdensity δ is defined as

$$\delta = \frac{n - \bar{n}}{\bar{n}}, \quad 12.$$

where n ($\bar{n}$) is the (average) number density of LAEs in a volume or an area. Because LAEs are low-mass galaxies (Section 5), LAEs show overdensity regions whose δ values are smaller than those of LBG and submillimeter galaxy (SMG) overdensity regions. However, overdensity regions of mass distribution are pinpointed with an LAE sample if the LAE sample is large enough to provide δ measurements with a high statistical accuracy distinguishing the overdensity regions from the field.

Theoretical models suggest that LAE overdensities with large δ values are progenitors of clusters, i.e., protoclusters (Chiang et al. 2013). Protoplusters of LAE overdensities are identified at $z \sim 2\text{--}7$ by blank-field surveys such as large-area NB observations (Ouchi et al. 2005, Yamada et al. 2012, Harikane et al. 2019). **Table 3** summarizes the spectroscopically confirmed protoclusters identified by the LAE blank-field surveys and compares protoclusters identified with the other types of galaxies such as LBGs and SMGs.

Table 3 List of protoclusters^a

Protocluster name	Redshift z	Number of member galaxies ^b N_{spec}	Overdensity ^c δ	Sample types ^d	Total mass ^e M_{total}	References
z66OD	6.59	12	14.3 ± 2.1	LAE	5.4×10^{14}	Harikane et al. 2019
HSC-z7PCC26	6.54	14	$6.8^{+6.1}_{-3.7}$	LAE	8.4×10^{14}	Chanchaiworawit et al. 2017, 2019; Higuchi et al. 2019
SDF	6.01	10	16 ± 7	LBG	$(2-4) \times 10^{14}$	Toshikawa et al. 2014
z57OD	5.69	44	11.5 ± 1.6	LAE	4.8×10^{14}	Ouchi et al. 2005, Jiang et al. 2018, Harikane et al. 2019
SPT2349-56	4.31	14	$>1,000$	SMG	1.16×10^{13}	Miller et al. 2018
TNJ1338-1942	4.11	37	$3.7^{+1.0}_{-0.8}$	LAE/LBG	$(6-9) \times 10^{14}$	Venemans et al. 2007, Miley et al. 2004, Zirm et al. 2005, Overzier et al. 2008
DRC-protocluster	4.00	10	$\sim 5.5-11.0$	SMG	$3.2-4.4 \times 10^{13}$	Oteo et al. 2018
PC217.96+32.3	3.79	65	14 ± 7	LAE	$(0.6-1.3) \times 10^{15}$	Lee et al. 2014, Day et al. 2016, Shi et al. 2019
D4GD01	3.67	11	NDA	LBG	NDA	Toshikawa et al. 2016
CJ0227-0421	3.29	19	10.5 ± 2.8	Spec	$(1.9-3.3) \times 10^{14}$	Lemaux et al. 2014
TNJ2009-3040	3.16	>11	$0.7^{+0.8}_{-0.6}$	LAE	NDA	Venemans et al. 2007
MRC0316-257	3.13	31	$2.3^{+0.5}_{-0.4}$	LAE	$(3-5) \times 10^{14}$	Venemans et al. 2007
SSA22FLD	3.09	>15	$3.6^{+1.4}_{-1.2}$	LBG/LAE/SMG	$(1.0-1.4) \times 10^{15}$	Steidel et al. 2000, Matsuda et al. 2005, Yamada et al. 2012, Umehata et al. 2018

Abbreviations: LAE, Lyman- α emitter; LBG, Lyman break galaxy; NDA, no data available; SMG, submillimeter galaxy; Spec, spectroscopically identified galaxy.

^aSpectroscopically confirmed protoclusters having >10 galaxies with spectroscopic redshifts. The protoclusters are listed in order of redshift. This table is the abstract of table 3 of Harikane et al. (2019).

^bConfirmed by spectroscopy.

^cNote that the length scales of the galaxy overdensity measurements differ.

^dTypes of member galaxies so far identified.

^eTotal (expected) mass of the overdensity at the redshift ($z = 0$) in units of solar masses.

Clustering properties of LAEs are quantified with correlation functions in two- or three-dimensional space. **Figure 16a** (subpanel *i*) presents the two-dimensional space correlation functions $\omega(\theta)$, i.e., the angular correlation functions (ACFs), of $\gtrsim L^*$ LAEs at $z = 5.7$ and 6.6 (Ouchi et al. 2018). Clearly, the ACFs of the LAEs are stronger than those of underlying dark matter $\omega_{\text{DM}}(\theta)$ predicted by the linear theory of the Λ CDM model (**Figure 16a**, *i*). In the Λ CDM framework, the excess of $\omega(\theta)$ above $\omega_{\text{DM}}(\theta)$ is evaluated by the bias b that is defined as

$$b^2 \equiv \frac{\omega(\theta)}{\omega_{\text{DM}}(\theta)}. \quad 13.$$

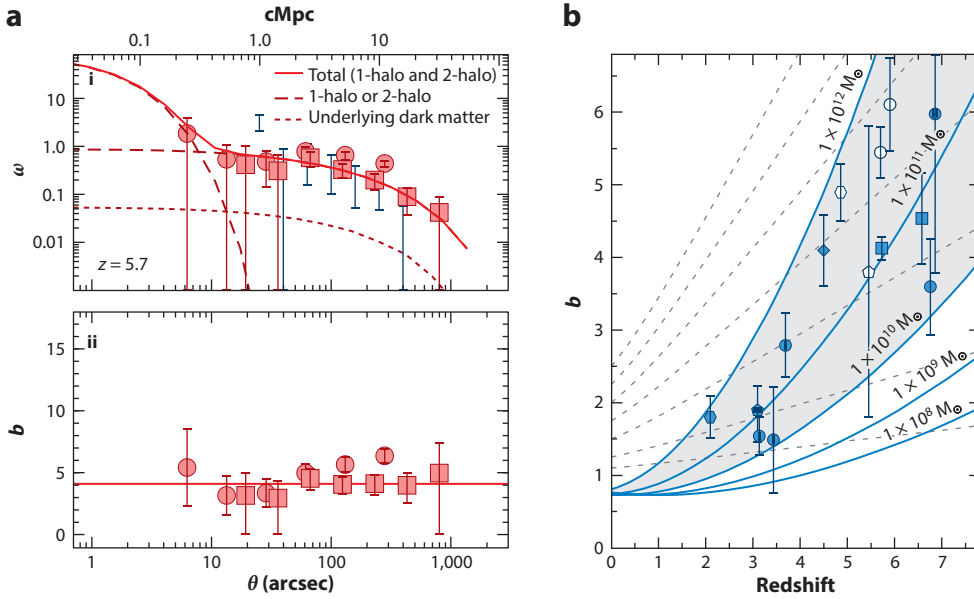


Figure 16

(*a, i*) ACF and (*a, ii*) bias of $\gtrsim L^*$ LAEs at $z = 5.7$. The data points indicate the (*a, i*) ACFs and (*a, ii*) bias of the LAEs. In subpanel *i*, the solid line presents the best-fit HOD model. The 1-halo and 2-halo terms of the best-fit HOD models are shown with the dashed lines that overlap with the solid line at the small and large scales, respectively. The dotted line denotes the underlying dark matter predicted by the linear theory. In subpanel *ii*, the horizontal line represents the average bias of the LAEs. The top axis denotes the projected distance in cMpc. (*b*) Bias of $\gtrsim L^*$ LAEs as a function of redshift. The data points represent the bias of the LAEs at $z = 2-7$. The solid lines indicate bias of dark matter halos with a dark matter halo mass of $10^8, 10^9, 10^{10}, 10^{11}$, and $10^{12} M_\odot$ in the case of one-to-one correspondence between galaxies and dark matter halos. The gray region shows the dark matter halo mass range of $10^{10}-10^{12} M_\odot$, where the biases of LAEs at $z = 2-7$ fall, whereas the number-weighted average mass is $10^{10}-10^{11} M_\odot$. The dotted lines are evolutionary tracks of bias in the case of the galaxy-conserving model. Panels *a* and *b* adapted from Ouchi et al. (2018) with permission. Abbreviations: ACF, angular correlation function; cMpc, comoving megaparsec; HOD, halo occupation distribution; LAEs, Lyman- α emitters.

The definition of the bias is also given by $b^2 \equiv \xi(r)/\xi_{\text{DM}}(r)$, where $\xi(r)$ and $\xi_{\text{DM}}(r)$ are the same as $\omega(\theta)$ and $\omega_{\text{DM}}(\theta)$, respectively, but for the three-dimensional space (spatial) correlation functions at the scale of r . **Figure 16a (subpanel ii)** shows bias of the LAEs as a function of angular distance. The bias of the LAEs evolves by redshift and increases toward high z (**Figure 16b**). The average bias of L^* LAEs is $b \sim 1-2$ at $z \sim 2-3$ and $b \sim 4-5$ at $z \sim 6-7$ (Ouchi et al. 2018). This increase of bias toward high z is explained by the physical picture that a galaxy forms only at a rare peak of density fluctuations, where dark matter halos are made, at the early stage of cosmic structure formation (Bardeen et al. 1986). Based on the average bias evolution of **Figure 16b**, the low bias values indicate that LAEs at $z \sim 2-3$ may be progenitors of today's MW-like galaxies (Gawiser et al. 2007), whereas the high bias values suggest that LAEs at $z \sim 5-7$ are progenitors of present-day massive elliptical galaxies (Ouchi et al. 2010).

There are large scatters in the bias measurements of LAEs obtained to date (see **Figure 16b**). The scatters would be larger than the statistical uncertainties. It is suggested that the large scatters are made by the sample variance, a.k.a. cosmic variance, originated by the small survey volumes (Kusakabe et al. 2018). Although the survey volumes of LAEs are generally small, 10^6 cMpc³ or

less, on-going Subaru HSC and HETDEX observations are providing the measurements of bias with negligibly small cosmic variance effects.

The correlation function is modeled by the power law or the halo occupation distribution (HOD) model. The power law is the empirical relation that has been used since the early measurements of local galaxy correlation functions were obtained (Totsuji & Kihara 1969). The HOD model is the parameterized model providing the relation between observed galaxies and their dark matter halo hosts of the Λ CDM structure formation (Cooray & Sheth 2002). The parameters of the HOD model define the occupation of galaxies in a dark matter halo as a function of mass, including the dark matter halo mass limit for hosting a galaxy and the power-law slope (and the scatter) of the occupation number depending on dark matter halo mass. Once the occupation of galaxies in a dark matter halo is determined, a correlation function and abundance of galaxies can be predicted with the HOD model. The correlation function consists of two components, 1-halo and 2-halo terms in small ($\lesssim 1$ cMpc) and large ($\gtrsim 1$ cMpc) scales, respectively (Figure 16a, i). The 1-halo term signal is originated from clustering of galaxies within one dark matter halo, whereas the 2-halo term signal is made by clustering of galaxies hosted by different dark matter halos. The correlation function and the abundance of galaxies, thus predicted, are compared with those of observational results, which determine the HOD model best-fit parameters. The HOD model with the best-fit parameters reveals properties of dark matter halos hosting LAEs and indicates that the average mass of dark matter halos $\langle M_h \rangle$ are moderately small, $\langle M_h \rangle = 10^{10} - 10^{11} M_\odot$ for the LAEs (Figure 16a, i; Ouchi et al. 2018). Because this is the average value of the dark matter halo masses, there should exist more LAEs with masses higher and lower than the average mass in overdensity (i.e., protocluster) and underdensity regions, respectively.

Once the mass of the hosting dark matter halos is constrained by HOD modeling, the number density of the host dark matter halos n_{DMH} can be estimated with the Λ CDM model. The n_{DMH} value can be compared with the number density of the observed LAEs $n_{\text{Ly}\alpha}$. The Ly α duty cycle of LAEs, $DC_{\text{Ly}\alpha}$ is defined as the ratio of $n_{\text{Ly}\alpha}$ to n_{DMH} ,

$$DC_{\text{Ly}\alpha} = \frac{n_{\text{Ly}\alpha}}{n_{\text{DMH}}}. \quad 14.$$

$DC_{\text{Ly}\alpha}$ is the fraction of the Ly α -emitting galaxies to the dark matter halos for a given mass. In the physical picture of LAEs, the Ly α duty cycle is determined by two effects, the intermittent star-formation activity and time-dependent Ly α escape. Observational results suggest that $DC_{\text{Ly}\alpha}$ is about 1% (Gawiser et al. 2007, Ouchi et al. 2010), which is comparable with the predictions of the numerical simulations, $DC_{\text{Ly}\alpha} \sim 1\text{--}10\%$ (Nagamine et al. 2010). As detailed in this section, clustering measurements of LAEs are not only useful to probe the cosmic structures but also to understand the physical properties of LAEs, host dark matter halo masses, and Ly α duty cycles.

9. COSMIC REIONIZATION AND LYMAN- α

Deep observations for galaxies have reached the EoR at $z \gtrsim 6$, when the neutral hydrogen of the IGM is ionized (Fan et al. 2006; Figure 17a). Ly α photons from a galaxy are scattered by the partly neutral hydrogen at the EoR. Ly α lines are redshifted from the systemic velocity of a galaxy by $\sim 100\text{--}200$ km s $^{-1}$ on average (Section 6), avoiding the extremely strong resonant scattering at the rest-frame 1216 Å in the partly neutral IGM. However, the redshifted Ly α is still scattered by the neutral hydrogen of the IGM via the Ly α damping wing (DW). Ly α DW is a Ly α profile of the long wavelength tail given by the natural broadening (originated by the quantum effect).

There is another important mechanism of Ly α scattering in the neutral IGM at the EoR that is illustrated in Figure 17. Theoretical models of reionization predict that UV radiation of a galaxy makes an ionized bubble around the galaxy in the neutral IGM and that the galaxy resides

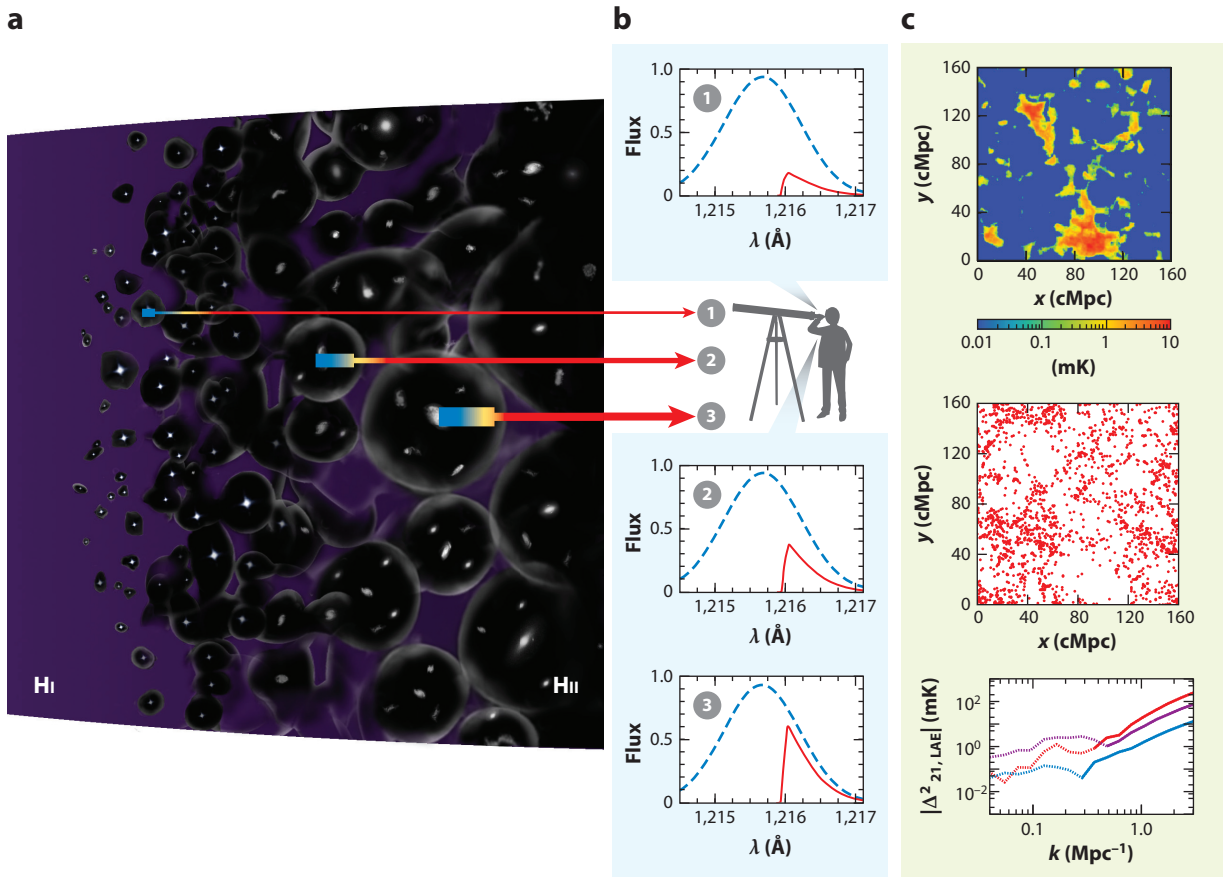


Figure 17

(a) Conceptual diagram of observations for LAEs at the epoch of reionization in the expanding Universe. The purple and black colors indicate the neutral and ionized IGM, respectively. LAEs in the ionized IGM (i.e., ionized bubbles) emit Ly α photons that are scattered by the neutral hydrogen IGM. The Ly α photons emitted in the large ionized bubbles can escape, being dimmed by a small amount of scattering, due to the Ly α velocity redshifted from the IGM velocity of the Hubble flow. In the diagram, wavelengths of Ly α emission are expressed with the spectral bands of the blue to red colors, and the widths of the spectral bands indicate Ly α emission intensities. (b) Model Ly α spectra (Dijkstra et al. 2007). The Ly α lines produced by galaxies are shown with blue dashed curves. The observed Ly α lines are shown with red curves. These galaxies reside at ionized bubbles whose radii are small (0 pMpc, i.e., fully neutral IGM; ①), medium (2 pMpc; ②), and large (10 pMpc; ③). (c) 21-cm brightness temperature map at $z = 6.6$ predicted by numerical simulations (top; Kubota et al. 2018). The blue regions of 0 mK correspond to the fully ionized regions. The middle panel shows positions of the model LAEs (red dots) in the cosmic volume that are the same as those in the top panel. The bottom panel presents LAE–21-cm cross-power spectra calculated on the basis of the models of the top and middle panels for three conditions of neutral hydrogen fractions, 1.7% (blue line), 31% (purple line), and 60% (red line). The solid and dotted lines indicate the positive and negative cross-power spectra, respectively. For display purposes, the signs of the negative cross-power spectra (dotted lines) are changed to the positive signs. The data of panels b and c are taken from Dijkstra et al. (2007) and Kubota et al. (2018), respectively, with permission. Abbreviations: IGM, intergalactic medium; LAEs, Lyman- α emitters; pMpc, physical megaparsec.

within the ionized bubble (Figure 17a). In this theoretical picture, Ly α photons can escape from the partly neutral IGM via the ionized bubble. At the back of the bubble on the border between the ionized and neutral hydrogen IGM, the neutral hydrogen gas is redshifted from the galaxy by the Hubble flow, which helps Ly α photons of the galaxy escape from the partly neutral IGM to the observer (Figure 17b).

Constraining the amount of the Ly α scattering in the IGM by observations, various studies have estimated the neutral hydrogen fraction x_{HI} or the ionized fraction Q_{HII} :

$$x_{\text{HI}} = \frac{n_{\text{HI}}}{(n_{\text{HI}} + n_{\text{HII}})}, \quad 15.$$

$$Q_{\text{HII}} = 1 - x_{\text{HI}}, \quad 16.$$

where n_{HI} and n_{HII} are the densities of neutral and ionized hydrogen, respectively. A volume-averaged x_{HI} (or Q_{HII}) is derived by Ly α studies.

Recent observational studies have identified signatures of Ly α photons scattered by Ly α DW of the neutral hydrogen gas at the EoR. Bright continuum objects, gamma-ray bursts (GRBs), and QSOs at the EoR are good probes of Ly α DW absorption with an intrinsic continuum spectrum modeled with a power law and an average spectrum of lower-redshift QSOs, respectively (Totani et al. 2016, Bañados et al. 2018). However, the small numbers of GRBs and quasars can probe the neutral hydrogen of the IGM on a small number of sight lines. For example, a theoretical model suggests that a single-object estimate of a GRB gives a systematic bias at the moderately high level of $\Delta x_{\text{HI}} \sim 0.3$ due to the patchy distribution of the neutral hydrogen at the EoR (McQuinn et al. 2008). Furthermore, there are well-known systematics of the ionization state of the IGM around a QSO (i.e., proximity effect; Bajtlik et al. 1988). Here, observations of LAEs can complement those of bright continuum objects, i.e., GRBs and QSOs. Although there are a variety of Ly α emission spectral shapes in LAEs, one can investigate a large number of LAEs distributed from high- to low-density regions. The large statistics of LAEs allow us to understand the evolution of the average neutral hydrogen fraction with negligible bias raised by the diversity of Ly α emission spectral shapes (Weinberger et al. 2019).

There are two popular techniques that study the average amount of Ly α DW absorption in the IGM with LAEs. One technique is the Ly α LF via the comparison with the UV-continuum LF (Malhotra & Rhoads 2004, Kashikawa et al. 2011, Itoh et al. 2018). Ly α and UV-continuum LDs are derived with the LFs. Note that the redshift evolution of the Ly α LD is determined by galaxy evolution and Ly α opacity of the IGM. Because the UV-continuum LD is a tracer of SFR and the dust extinction in the ISM, the fact that Ly α LD decreases faster than the UV-continuum LD evolution suggests an increase of Ly α opacity of the IGM, which is shown in **Figure 7a** (see Itoh et al. 2018, their figure 11, for statistically significant results). Ly α LD rapidly drops from $z \sim 6.5$ to 7.5, whereas the evolution of the UV-continuum LD is milder than that of the Ly α LD. This evolutionary difference allows us to estimate the Ly α opacity via the comparisons with analytical and numerical models of cosmic reionization (Furlanetto et al. 2006, McQuinn et al. 2007). Another technique is to investigate a fraction of Ly α -emitting galaxies out of all of the UV-continuum-selected galaxies (Pentericci et al. 2011, Stark et al. 2011, Ono et al. 2012). **Figure 18a** and **Table 4** summarize the measurements of the Ly α -emitting galaxy fractions for galaxies with the two different limits of the UV-continuum magnitudes. In **Figure 18a** and **Table 4**, we show the average Ly α -emitting galaxy fractions with standard errors, which are estimated from the compilation of the data so far obtained by various observational studies (Stark et al. 2011, Curtis-Lake et al. 2012, Mallery et al. 2012, Ono et al. 2012, Treu et al. 2013, Schenker et al. 2014, Tilvi et al. 2014, Cassata et al. 2015, De Barros et al. 2017, Arrabal Haro et al. 2018, Caruana et al. 2018, Pentericci et al. 2018, Mason et al. 2019, Kusakabe et al. 2020). The Ly α -emitting galaxy fraction increases from $z \sim 4$ to 6, whereas this fraction clearly decreases from $z \sim 6$ to 8, indicating that the Ly α opacity increases from $z \sim 6$ to 8. The decrease of the Ly α -emitting galaxy fraction is modeled, providing estimates of the cosmic average x_{HI} . The x_{HI} estimates, thus obtained, are summarized in **Figure 18b**,

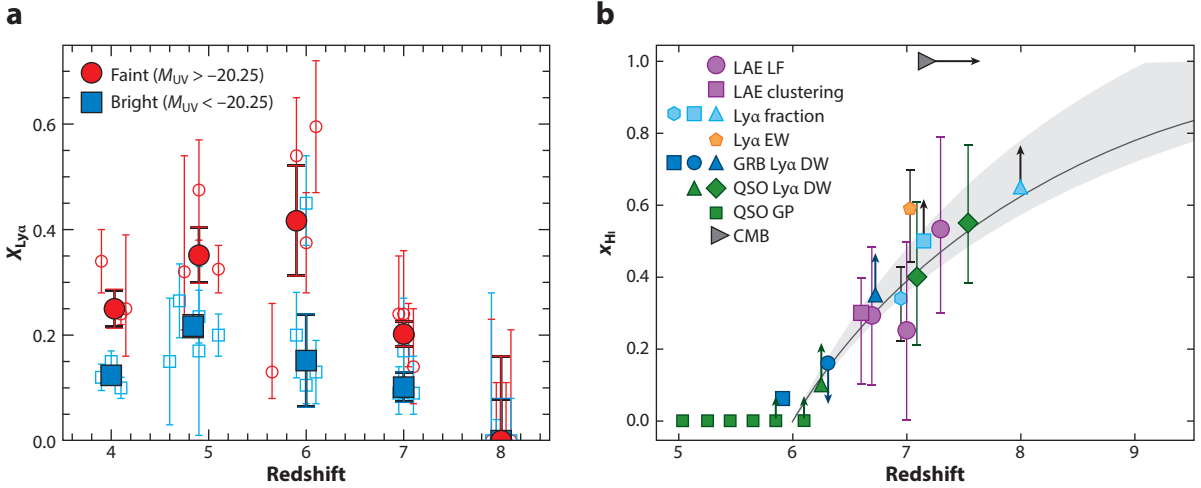


Figure 18

(a) Fraction of LAEs with $\text{Ly}\alpha\text{ }EW_0 \geq 25\text{ }\text{\AA}$, $X_{\text{Ly}\alpha}$, as a function of redshift. The red-filled circles and the blue-filled squares indicate the average $X_{\text{Ly}\alpha}$ values for galaxies with the faint ($M_{\text{UV}} > -20.25$) and bright ($M_{\text{UV}} < -20.25$) UV-continuum magnitudes, respectively. These average $X_{\text{Ly}\alpha}$ values are estimated with the compilation data of the red open circles ($M_{\text{UV}} > -20.25$) and the blue open squares ($M_{\text{UV}} < -20.25$) that are obtained by many observational studies (Stark et al. 2011, Curtis-Lake et al. 2012, Mallery et al. 2012, Ono et al. 2012, Treu et al. 2013, Schenker et al. 2014, Tilvi et al. 2014, Cassata et al. 2015, De Barros et al. 2017, Arrabal Haro et al. 2018, Caruana et al. 2018, Pentericci et al. 2018, Mason et al. 2019, Kusakabe et al. 2020). (b) Neutral hydrogen fraction as a function of redshift (Itoh et al. 2018). The purple circles represent x_{HI} estimates given by the $\text{Ly}\alpha$ luminosity density evolution obtained with luminosity function measurements. The cyan hexagon, square, and triangle are the constraints of x_{HI} given by the $\text{Ly}\alpha$ -emitting galaxy fraction measurements. The orange pentagon shows the x_{HI} estimate calculated with the $\text{Ly}\alpha$ EW distribution of dropout galaxies. The purple square denotes the x_{HI} value estimated with the LAE clustering. The blue triangle, square, and circle indicate x_{HI} values given by GRB $\text{Ly}\alpha$ DW measurements. The green diamonds and triangle show x_{HI} constraints obtained by the QSO $\text{Ly}\alpha$ DW observations, while the green squares represent those of QSO GP optical depths. See Itoh et al. (2018, their figure 14) for the references for the measurements. The dark gray triangle presents the 1σ lower limit obtained with the CMB Thomson scattering optical depth measurement. The black curve and the gray shading denote the best-estimate x_{HI} and the error given with Equations 17 and 21 and galaxy ρ_{UV} measurements (Ishigaki et al. 2018). Panel b adapted from Itoh et al. (2018) with permission. Abbreviations: CMB, cosmic microwave background; DW, damping wing; EW_0 , rest-frame equivalent width; GP, Gunn–Peterson; GRB, gamma-ray burst; LAE, Lyman- α emitter; QSO, quasi-stellar object.

Table 4 Fraction of $\text{Ly}\alpha$ -emitting galaxies with $EW_0 \geq 25\text{ }\text{\AA}$

Sample	Magnitude	Redshift	$X_{\text{Ly}\alpha}$
Bright	$M_{\text{UV}} < -20.5$	4.0	0.12 ± 0.02
		4.8	0.22 ± 0.02
		6.0	0.15 ± 0.09
		7.0	0.10 ± 0.03
		8.0	<0.08
Faint	$M_{\text{UV}} > -20.5$	4.0	0.25 ± 0.04
		4.9	0.35 ± 0.05
		5.9	0.42 ± 0.10
		7.0	0.20 ± 0.02
		8.0	<0.16

indicating the cosmic reionization history. The neutral hydrogen gas fraction increases from $z \sim 6$ toward high z , and the mid-point of cosmic reionization, $x_{\text{HI}} = 50\%$, is estimated to be $z \sim 7.5$.

$\text{Ly}\alpha$ emission is the strongest line found in the optical wavelength for the majority of objects at $z \sim 2\text{--}7$. Although this characteristic of $\text{Ly}\alpha$ emission is very advantageous for studies of high- z objects, this advantage may not be true for sources at the middle or early EoR. This is because the chance of a detection of a $\text{Ly}\alpha$ -emitting galaxy decreases toward high z (**Figure 18a**), due to the fact that $\text{Ly}\alpha$ photons from galaxies are scattered by the neutral hydrogen of the IGM at the EoR $z > 6$. If LAEs and the IGM are static and uniformly distributed, no $\text{Ly}\alpha$ photons can escape from the neutral Universe. However, in reality, $\text{Ly}\alpha$ photons come out of the highly neutral IGM with the help of the $\text{Ly}\alpha$ velocity shifts (from the neutral IGM) given by the peculiar motions of galaxies as well as by the ionized bubbles around clustered galaxies. Theoretical models indicate that about 10% of $\text{Ly}\alpha$ fluxes can escape from the highly neutral IGM by the help of these physical effects (Gnedin & Prada 2004). There is an open question whether a redshift limit of a $\text{Ly}\alpha$ detection exists and what redshift is the limit, if it exists. A future accurate measurement of the $\text{Ly}\alpha$ observability at the early EoR ($z \sim 10$) will allow us to understand the physical properties of the LAE dynamics and the ionized bubbles.

Theoretical studies predict that LAEs are key sources for identifying signals of the H I 21-cm emission originated from neutral hydrogen at the EoR (Lidz et al. 2009, Sobacchi & Mesinger 2015, Hutter et al. 2018). Although there are several high-sensitivity radio telescopes targeting the EoR 21-cm emission (e.g., LOw Frequency ARray or LOFAR; Jelić et al. 2014), no signals of the EoR 21-cm emission have been detected so far, mainly because of the bright foreground emission of Earth's ionosphere. Here, LAEs can be used as signposts for the EoR 21-cm emission detections. The large sample of LAEs with known positions and redshifts allows us to conduct cross-correlation analysis with EoR 21-cm data (Lidz et al. 2009), and to identify a signal of the EoR 21-cm emission, removing the problematic foreground emission. The bottom subpanel of **Figure 17c** presents the LAE–21-cm cross-power spectra predicted by numerical simulations (Kubota et al. 2018). These cross-power spectra are calculated with the simulation data of the 21-cm brightness temperature (LAE distribution) map shown in the top (middle) subpanel of **Figure 17c**. The cross-power spectrum consists of two components at small and large scales. The large-scale component is the negative signal (i.e., anticorrelation) representing ionizing sources (LAEs) that make ionized bubbles (i.e., no neutral hydrogen around LAEs), whereas the small-scale component is the positive signal indicating that LAEs form in an overdensity of neutral hydrogen. There exists a transition between the negative and positive signals, which are due to the ionized bubbles around LAEs. Detections of the negative and positive signals will confirm an existence of ionized bubbles at the EoR. In summary, the LAE–21-cm cross-correlation analysis will help the detections of the EoR 21-cm signals and characterize the cosmic reionization process via the ionized bubble topology.

LAEs are young SFGs that have abundant massive stars producing ionizing photons contributing to cosmic reionization. For faint SFGs with $M_{\text{UV}} > -20.25$, the number fraction of LAEs reaches $\sim 40\%$ of the faint SFGs at the redshift of the EoR $z \sim 6^{16}$ (**Figure 18a**), indicating that LAEs are a major population of SFGs that supply ionizing photons for cosmic reionization. In fact, the UV-continuum LF of LAEs is comparable with that of LBGs at $z \sim 6$, whereas those of LAEs fall below those of LBGs by an order of magnitude at $z \sim 3\text{--}4$ (see Ouchi et al. 2008, their figure 22).

¹⁶Although the fraction of LAEs decreases from $z \sim 6$ toward high z , this decrease is just made by a low observability given by the external effect of the neutral IGM that scatters $\text{Ly}\alpha$ photons.

With the ionizing photon production rate $\dot{n}_{\text{ion}}$ that is defined by the number of ionizing photons per volume and time, the budget of ionizing photons in cosmic reionization is modeled in the one-zone model of the ionization equation (Madau et al. 1999, Robertson et al. 2015). Here the derivative of ionized fraction (Equation 16) with respect to time $\dot{Q}_{\text{HII}}$ is written as

$$\dot{Q}_{\text{HII}} = \frac{\dot{n}_{\text{ion}}}{\langle n_{\text{H}} \rangle} - \frac{Q_{\text{HII}}}{t_{\text{rec}}}, \quad 17.$$

where $\langle n_{\text{H}} \rangle$ and t_{rec} are the average hydrogen number density and the recombination time, respectively, defined by

$$\langle n_{\text{H}} \rangle = \frac{X_{\text{p}} \Omega_{\text{b}} \rho_{\text{c}}}{m_{\text{H}}}, \quad 18.$$

$$t_{\text{rec}} = \frac{1}{C_{\text{HII}} \alpha_{\text{B}}(T) (1 + Y_{\text{p}}/4X_{\text{p}}) \langle n_{\text{H}} \rangle (1 + z)^3}. \quad 19.$$

The parameters, m_{H} , ρ_{c} , and X_{p} (Y_{p}) are the mass of the hydrogen atom, the critical density, and the primordial mass fraction of hydrogen (helium), respectively. In Equation 19, the parameters of C_{HII} and $\alpha_{\text{B}}(T)$ represent the clumping factor and the case B hydrogen recombination coefficient for the IGM temperature T at a mean density, respectively. All of these parameters are determined by physics and cosmology, except for C_{HII} and $\dot{n}_{\text{ion}}$. The parameter of $\dot{n}_{\text{ion}}$ is closely related to galaxy formation, and estimated by

$$\dot{n}_{\text{ion}} = \int_0^{+\infty} f_{\text{esc}}^{\text{ion}}(L_{\text{UV}}) \xi_{\text{ion}}(L_{\text{UV}}) \phi(L_{\text{UV}}) L_{\text{UV}} dL_{\text{UV}}, \quad 20.$$

$$= f_{\text{esc}}^{\text{ion}} \xi_{\text{ion}} \rho_{\text{UV}} \quad [\text{for no } L_{\text{UV}} \text{ dependences}], \quad 21.$$

where $f_{\text{esc}}^{\text{ion}}$ and ξ_{ion} are the ionizing photon escape fraction¹⁷ and the ionizing photon production efficiency, respectively. The values of L_{UV} , $\phi(L_{\text{UV}})$, and ρ_{UV} are the UV-continuum luminosity, the UV-continuum LF, and the UV-continuum LD, respectively (Section 3). Equation 21 assumes no luminosity dependence in the parameters of $f_{\text{esc}}^{\text{ion}}$ and ξ_{ion} .

There are three major parameters for $\dot{n}_{\text{ion}}$: (a) ρ_{UV} , (b) ξ_{ion} , and (c) $f_{\text{esc}}^{\text{ion}}$. (a) The LAEs' contribution to ρ_{UV} is large at high redshift. As discussed in Section 3, the fraction of LAEs to all galaxies in ρ_{UV} increases from the local Universe toward the EoR $z \sim 6$. This increase indicates that LAEs' contribution to the cosmic SFR density increases (Ouchi et al. 2008, Ciardullo et al. 2012). Because LAEs are abundant and major galaxies at the EoR, ionizing photon emission properties of LAEs are critical to understanding sources of cosmic reionization. (b) For photon production, it was recently claimed that the LAEs' ξ_{ion} is higher than other high- z galaxy populations. The typical ionizing photon production efficiency of LAEs is $\log(\xi_{\text{ion}}/[\text{Hz erg}^{-1}]) \sim 25.5$, which is significantly higher than those of LBGs at a given UV magnitude (Harikane et al. 2018). (c) It is suggested that the average ionizing photon escape fraction is also high for LAEs. Deep spectroscopy and NB imaging observations reveal that the average $f_{\text{esc}}^{\text{ion}}$ value is $\sim 20\%$ of LAEs at $z \sim 3$ (Shapley et al. 2006; see also de Barros et al. 2016, Vanzella et al. 2016). In other words, a positive correlation between $f_{\text{esc}}^{\text{ion}}$ and $\text{Ly}\alpha \text{ EW}_0$ is suggested (Iwata et al. 2009, Nestor et al. 2013).¹⁸ Deep NIR spectroscopy studies find that the line flux ratios of $[\text{OIII}] \ 5007$ to $[\text{OII}] \ 3727$,

¹⁷Note that this parameter of $f_{\text{esc}}^{\text{ion}}$ is different from the escape fraction of $\text{Ly}\alpha$ photons (Equation 9).

¹⁸Because $\text{Ly}\alpha$ emission is produced by recombination in hydrogen gas of the ISM that absorbs ionizing photons, strong $\text{Ly}\alpha$ emission should not be found in galaxies emitting ionizing photons. However, even under

O32 (Section 6.5), of LAEs at $z \sim 2$ are large, O32 ~ 10 (Nakajima et al. 2013). Because such galaxies with large O_{32} values show a high ionizing photon escape fraction, $f_{\text{esc}}^{\text{ion}} \sim 5\text{--}50\%$ (Izotov et al. 2016, Vanzella et al. 2016), LAEs are thought to be major emitters of ionizing photons at high redshift. For the physical origin of ionizing photon emission of LAEs, various scenarios including density-bounded nebulae are suggested (Nakajima & Ouchi 2014). However, the physical reason for escaping ionizing photons is still under debate.

10. OPEN QUESTIONS AND FUTURE PROSPECTS

In observational studies of Ly α emission, there are five major questions,¹⁹ which are listed below.

1. What are the dominant sources of Ly α photons in LAEs that include ELANe, LABs, and LAHs? Are the majority of Ly α photons produced in HII regions of star formation and/or AGNs? Are there any major contributions of Ly α photons from gravitational cooling? (Section 2.)
2. How faint can the luminosities of LAEs be? What is the dark matter low-mass limit for SFGs that emit Ly α ? What is the faint-end slope α of the Ly α LF? (Section 3.)
3. What is the major physical origin of the high Ly α emissivity of LAEs such as those characterized by the large Ly α EW_0 and the high $f_{\text{esc}}^{\text{Ly}\alpha}$? What makes LAEs different from other SFGs with weak-to-no Ly α emission? (Sections 3–8.)
4. What are the ionization states of the ISM and CGM of LAEs? Is the density-bounded picture possible in the ISM? Do regular LAEs (not including ELANe with an active AGN) have CGM gas that mainly consists of neutral hydrogen? What is the cause of the observational trend of the high fraction of Lyman-continuum leaking for LAEs that is important for sources of reionization? (Sections 6, 7, and 9.)
5. Do Pop III LAEs exist as originally predicted by Partridge & Peebles (1967)? If Pop III LAEs exist, are they observable at the post EoR (late epoch) and/or the EoR (early epoch), which may or may not have a mechanism allowing the Ly α escape from the partly neutral Universe? Does the Ly α escape from the partly neutral Universe depend on reionization topology? (Sections 6 and 9.)

There are a number of ongoing and next-generation instrument projects that are critical for resolving these five questions. Such instruments include those for statistical studies [Subaru/HSC, Prime-Focus Spectrograph (PFS), and HETDEX], high sensitivity observations (VLT/MUSE and Keck/KCWI; Morrissey et al. 2018), and submillimeter/millimeter wavelength observations (ALMA). Ly α emission studies will be boosted by the great sensitivities of the next-generation large telescopes, the *James Webb Space Telescope* (JWST) and extremely large telescopes (ELTs). Furthermore, the H I 21-cm observations with the Hydrogen Epoch of Reionization Array (HERA) and the Square Kilometer Array (SKA) will complement the Ly α emission studies at the EoR (Section 9). Because future imaging and spectroscopic data such as that taken with HSC, PFS, and HETDEX are too large to be analyzed by visual inspections for quality assessments, one will need the new analysis technique of artificial intelligence that includes machine learning.

In the past two decades, observations have targeted SFGs emitting Ly α that are easily detected. Recent studies change the observational targets to diffuse Ly α emission around galaxies

this physical picture, model calculations suggest that there is a positive correlation between $f_{\text{esc}}^{\text{ion}}$ and Ly α EW_0 in the regime of the moderately low Ly α EW_0 values (Nakajima & Ouchi 2014).

¹⁹In addition to these five major questions, various observational measurements should be refined and revisited. One example is the determination of Ly α LFs, where the flux contributions of extended LAHs are not included in many of the previous studies (Herenz et al. 2019).

and filaments of the large-scale structures that are new observational frontiers. Although Ly α emission traces both HII and HI gas with great sensitivity, the origins of the Ly α emission cannot be clearly distinguished between, e.g., recombination and resonant scattering. The future observing programs should combine the Ly α emission results with the complementary nonresonant hydrogen emission (H α and H β) and HI absorption observations (Lee et al. 2014), exploiting the high-sensitivity instruments and the next-generation large telescopes. Including the 21-cm observations at the EoR, future observations will reveal the HI and HII gas distribution in the Universe in the scales from star-forming regions to the large-scale structures.

11. SUMMARY AND FINAL THOUGHTS

Since the first discoveries of strong Ly α -emitting high- z galaxies in the late 1990s, Ly α emission has been one of the major probes for the high-redshift Universe. Defining LAEs as galaxies with Ly α $EW_0 \gtrsim 20$ Å, more than 1,000 (20,000) LAEs are spectroscopically (photometrically) identified to date. Observations suggest that LAEs emerge at high redshift, and the fraction of LAEs to the other galaxies increases (for a given UV-continuum luminosity) toward high redshift up to the EoR (Section 3). LAEs are galaxies of a major population at high redshift. This emergence of LAEs is understood by the increase of Ly α escape fraction for high- z SFGs.

At high redshift ($z \gtrsim 2$), a majority of LAEs with a L^* Ly α luminosity are low mass ($M_* \sim 10^8$ – $10^9 M_\odot$ and $M_h \sim 10^{10}$ – $10^{11} M_\odot$) galaxies with star-forming activity (1 – $10 M_\odot \text{ year}^{-1}$; Figure 9). Such typical LAEs are high- z analogs of dwarf galaxies having a young dust-poor stellar population with a subsolar metallicity in the ISM gas, showing a compact disk morphology ($n_s \sim 1$) with an effective radius of ~ 1 pkpc (Sections 4–6). These physical properties of LAEs can be explained by the nature of the faint continuum (sub L_{UV}^* ; Section 3). In fact, LAEs follow the size–mass relation and the SFR–mass metallicity relation. LAEs are probably SFGs similar to those of dropout galaxies or LBGs that have weak or no Ly α emission. In this case, the difference of LAEs from the other galaxies is characterized by the Ly α emission duty cycle of $\sim 1\%$ (Section 8) caused by the combination of intermittent star formation and time-dependent Ly α escape.

Ly α emission is not only observed at the ionized hydrogen clouds by recombination but also at the neutral hydrogen clouds by resonant scattering. The resonant nature of Ly α allows us to probe diffuse gas in the ISM and/or CGM of a galaxy as well as the IGM. For the ISM, the skewed profiles of galaxy Ly α emission suggest the existence of neutral hydrogen gas outflowing from Ly α sources (such as star-forming regions), which is explained by the ES model (Sections 2 and 6). For the CGM (Section 7), diffuse Ly α emission around an SFG, i.e., LAH, is ubiquitous. Although it is not understood whether the LAH is made by the recombination process of the ionized clouds or the resonant scattering of the neutral clouds, the CGM includes metal up to at least ~ 10 pkpc such as that identified by the extended carbon emission with a radial profile similar to those of LAHs, which is indicative that the CGM is metal enriched by outflow. The LAH extends beyond the virial radius of a galaxy dark matter halo up to 1 cMpc or more, perhaps originating from filaments in the large-scale structures.

Strong Ly α emission is easily detected by observations, being used as a signpost of high- z objects. Reports of Ly α emission detection have been made for galaxies up to $z \sim 9$ to date. The redshift frontier of galaxy observations has entered into the EoR, where Ly α is scattered by the partly neutral hydrogen IGM. Large statistics of galaxy Ly α emission provides the average amount of Ly α scattered by the neutral IGM at the EoR, having revealed the cosmic reionization history, so far, up to the middle point of reionization $x_{\text{HI}} \sim 50\%$ at $z \sim 7$ – 8 (Section 9). The Ly α escape from the partly neutral hydrogen IGM is not only a probe of cosmic reionization history but also a signal of the galaxies and ionized-bubble relation (determined by galaxy clustering,

star-formation duty cycle, and galaxy peculiar motions). One critical question about Ly α sources is how early epoch, e.g., $z \sim 10$ or $z \sim 15$, Ly α emission can escape (Section 10).

Throughout this review, we have explained the observational differences between LAEs and other SFGs emitting weak-to-no Ly α , discussing possible physical origins of the high Ly α emissivity of LAEs such as that characterized by the large Ly α EW_0 and the high $f_{\text{esc}}^{\text{Ly}\alpha}$. Although the major physical origin of the Ly α emissivity difference is an open question as raised in item 3 of Section 10, we have a physical picture that a column density of H I gas toward a Ly α source (such as star-forming regions) is a key physical quantity that controls the Ly α emissivity. For LAEs, the Ly α offset velocities are comparable with outflow velocities ($\Delta v_{\text{Ly}\alpha} \sim V_{\text{exp}} \sim 200 \text{ km s}^{-1}$), and this kinematic property of LAEs can be explained by a small H I column density producing no significant backscattered Ly α photons (Section 6). The high O32 ratios and the existence of the strong high ionization lines suggest that the ISM of LAEs is highly ionized, and this photoionization property of LAEs indicates a small fraction of H I gas in LAEs (Section 6). There is an anticorrelation between Ly α EW_0 and the Ly α spatial offset $\delta_{\text{Ly}\alpha}$ on the sky, and this Ly α morphological property can be explained by the long mean-free path (less resonant scattering) of Ly α photons for LAEs with a small H I column density toward the observer (Section 4). This explanation for the Ly α EW_0 - $\delta_{\text{Ly}\alpha}$ anticorrelation is consistent with the fact that the LAHs are small for LAEs whose central part ($r < 8 \text{ pkpc}$) of Ly α luminosity is bright (Section 7). In this way, all of the independent observational properties of kinematics, photoionization, and morphology consistently support the physical picture that a column density of H I gas is a key physical quantity for the Ly α emissivity, and that LAEs are different from the other SFGs emitting weak-to-no Ly α due to the small H I column density of LAEs. In this physical picture, we would witness SFGs at the galaxy evolutionary phase of the highly ionized H I poor state as LAEs, when the SFGs are free from dusty and gaseous starbursts such as those triggered by gas-rich mergers. Because the cosmic average of $f_{\text{esc}}^{\text{Ly}\alpha}$ (i.e., average Ly α emissivity) increases from $z = 0$ to 6 by two orders of magnitude (Section 6), in galaxy formation a number of early young galaxies should experience the highly ionized H I poor (LAE) phase.

As reviewed in this article, observations of Ly α emission have driven various studies of the high-redshift Universe for more than two decades. Ly α observations have been pioneering the redshift frontier and investigating low-mass high- z galaxies. Furthermore, recently, Ly α observations have attracted attention for studying cosmic structures traced by diffuse neutral and ionized hydrogen gas of the CGM and the IGM, serving as a new observational role. Because the studies of the CGM and the IGM have been (will be) conducted by hydrogen and metal absorption observations (radio low-frequency 21-cm observations), forthcoming studies combining these absorption (radio) observations with Ly α emission from neutral and ionized hydrogen will open up new territory to study, that of cosmic structures and their connection to galaxy formation, which has remained a mystery to date.

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